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Evaluating Transformer-Based Decision Support for NHS Mental Health Triage: High-Risk Patient Identification and Prioritisation

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Declaration

I hereby declare that this dissertation is submitted in partial fulfilment of the requirements for the degree of BSc (Hons) in Computer Science at the University of Gloucestershire. To the best of my knowledge and belief, this dissertation has not been submitted, either in whole or in part, for any other degree or qualification at this or any other university.

Except where explicitly stated by reference or acknowledgment, all the work presented in this dissertation is entirely my own.

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Abstract

NHS mental health services face increasing pressure from rising demand and constrained resources, heightening the risk of delayed identification of patients at risk of self-harm. Current triage workflows rely heavily on manual interpretation of referral text, which is time-consuming, variable, and difficult to scale. This dissertation investigates whether transformer-based Natural Language Processing (NLP) can support safe and effective risk-based prioritisation of mental health referrals within a clinically governed decision-support framework.

A baseline TF-IDF and Logistic Regression model was extended into a fine-tuned BERT transformer architecture, trained and evaluated on 5,000 synthetic referral-style clinical records. The enhanced model achieved a substantial improvement in classification performance, with accuracy increasing from 48.80% to 70.80% and F1-score from 0.4635 to 0.6790, demonstrating the effectiveness of contextual embeddings in capturing nuanced clinical intent.

However, experimental results revealed a critical limitation: improved classification performance did not translate into effective prioritisation. The model failed to surface high-risk patients within the top decile of the triage queue (Recall@10% = 0.0000), exposing a structural disconnect between classification-optimised objectives and ranking-based clinical requirements. Further analysis identified optimisation constraints under resource-limited conditions, where aggressive batch scaling reduced gradient update frequency and led to convergence failure. Although hyperparameter rebalancing restored classification accuracy, ranking behaviour remained inadequate due to poorly calibrated and insufficiently discriminative probability outputs.

To address clinical safety, the model was integrated into a full-stack prototype comprising a FastAPI backend, React-based clinician interface, Explainable AI outputs, and a Human-in-the-Loop MLOps replay buffer for continual learning. The system is therefore positioned as a governance-aware decision-support tool rather than an autonomous triage mechanism.

This work demonstrates that while transformer-based NLP models significantly improve risk classification, safe deployment in mental health triage requires explicit alignment between probability calibration, ranking behaviour, and clinical oversight. It further establishes that classification metrics alone are insufficient for evaluating triage systems, and that ranking-aware evaluation must be treated as a first-class requirement in safety-critical NLP applications.

Ethical Issues Statement

This study involved no human participants and no real patient data. All experiments were conducted using a fully synthetic dataset generated under controlled conditions, ensuring complete anonymisation and compliance with GDPR and NHS data governance principles. The project was limited to a simulated clinical environment and did not involve access to NHS systems or recruitment of participants. As confirmed by the University of Gloucestershire research ethics checklist, submitted prior to the commencement of this research, the study raised no ethical risks requiring further review. Potential limitations relating to model bias, misclassification, and the use of synthetic data are acknowledged and discussed within the dissertation.

Dedicated to those working in mental health services.

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Chapter 1

Introduction

1.1 Overview

Mental health services in the NHS face unprecedented demand, with increasing numbers of patients requiring urgent assessment and intervention (Care Quality Commission (CQC), 2025). Timely identification of patients at high risk of self-harm is critical to patient safety and outcomes. Yet current triage processes are often manual, time-consuming, and subject to inconsistency.

NHS mental health triage teams process large volumes of referrals under time pressure, increasing the risk that subtle indicators of self-harm are overlooked. Advanced NLP algorithms can offer a consistent, data-driven approach to flagging patients at highest risk, thereby supporting more efficient and equitable triage. While NLP-based methods for risk detection have been validated across a range of healthcare contexts, relatively few studies have focused on improving existing algorithms and evaluating their deployment within simulated NHS workflows (Klug et al., 2024). This dissertation addresses that gap through model enhancement, prototype development, and evaluation that reflects real-world constraints and clinical decision-making processes. By exploring how transformer-based Large Language Models (LLMs) and Unified Textual Encoding can enhance the automated detection of self-harm risk indicators, this work contributes both methodologically and practically to the growing field of clinical decision support. It also examines whether improvements in classification performance translate into effective prioritisation behaviour within triage workflows, where ranking quality and probability calibration determine real-world clinical utility.

Beyond the technical focus, this research is motivated by the practical realities of frontline services. Any proposed system must be usable by clinicians who may not have specialist technical training, must integrate with existing NHS workflows, and must respect the sensitivities of working with vulnerable patients. The introduction therefore not only frames a computational challenge, but also sets out a broader question of how Deep Learning tools can be designed in a way that genuinely supports, rather than disrupts, everyday clinical practice.

1.2 Problem Statement

Current manual triage relies on the subjective interpretation of free text, making it vulnerable to cognitive fatigue and increasing the risk that severe cases are erroneously deprioritised (Alobayli, O'Connor, Holloway, and Cresswell, 2023). Existing NLP models address this inconsistency in theory but are rarely evaluated within real-world clinical constraints. A critical gap persists between isolated deep-learning prototypes optimised for classification accuracy and

operationally reliable decision-support tools that require well-calibrated probability outputs, effective prioritisation behaviour, explainability, and clinician trust (Overgaard et al., 2023).

1.3 Research Questions

This research aims to address the following questions:

1. To what extent can Natural Language Processing (NLP) models accurately identify indicators of self-harm risk within mental health referral-style text?
2. How does the use of LLM-generated synthetic data affect the performance and generalisability of NLP models for patient prioritisation tasks?
3. How can the performance, generalisability, and practical implications of the proposed model be effectively evaluated within a simulated NHS triage scenario?

1.4 Research Objectives

The objectives of this research are as follows:

1. To investigate current approaches to prioritising mental health patients within NHS triage systems and identify their limitations.
2. To develop an NLP-based classification model and an LLM-based support architecture for detecting indicators of self-harm risk in referral-style text.
3. To evaluate the performance and generalisability of the proposed model within a simulated NHS triage scenario.

1.5 Scope

This research focuses specifically on benchmarking and enhancing NLP-driven risk assessment frameworks for patient prioritisation in NHS mental health triage, demonstrated through the automated analysis of referral text for self-harm indicators. The approach addresses the challenges of clinical decision support in a high-stakes, data-rich environment and evaluates the practical reliability of a prototype system within simulated NHS workflows.

Because direct integration and testing against live NHS systems is not feasible within the constraints of an undergraduate project, this study instead constructs a simulated, full-stack triage environment. Referral-style text is generated and subsequently processed in a way that mimics NHS workflows, allowing the technical machine learning architecture to be evaluated while strictly respecting data boundaries, ethical constraints, and patient safety.

1.6 Ethical Considerations

Key ethical considerations include the following:

- Patient safety risks arising from false-negative predictions that may fail to flag high-risk patients, or false positives that may unnecessarily exhaust triage resources.

- Transparency and explainability of model outputs (XAI) to ensure clinicians understand how risk assessments are mathematically generated.
- Bias and algorithm performance scaling across diverse patient demographic groups.
- Data privacy and confidentiality addressed through the strict use of completely synthetic referral text, acting as a proxy for real records to adhere strictly to NHS data governance and GDPR requirements.
- Automation bias mitigation by securely designing the tool as a Decision Support System requiring Human-in-the-Loop oversight, rather than assigning autonomous decision replacement.

1.7 Conclusion and Dissertation Outline

This chapter has outlined the urgent clinical need for improved mental health triage and proposed an LLM-driven, transformer-based NLP framework to address this gap. The remainder of this dissertation is structured as follows:

Chapter 2 presents a comprehensive literature review, examining existing NLP applications across diverse domains, the use of LLMs in NHS mental health contexts, and identifying key methodological gaps. Chapter 3 outlines the research methodology and overall experimental framework for controlled evaluation. Chapter 4 explores the system design, detailing the underlying architecture, Unified Textual Encoding, and the end-to-end clinical interface workflow. Chapter 5 describes the deep learning implementation, including synthetic data generation and the model development pipeline within an active-learning MLOps framework. Chapter 6 presents and analyses the evaluation results, including hyperparameter convergence, model performance, and limitations in probability calibration. Chapter 7 concludes with reflections on the broader implications for LLM deployment in mental health services and provides recommendations for future research and implementation.

Chapter 2

Literature Review

This chapter reviews the current state of Deep Learning and Natural Language Processing (NLP) in healthcare decision support, focusing on mental health triage and self-harm risk detection. It critically examines technical benchmarks, existing NHS initiatives, key ethical considerations, and identifies gaps motivating this study.

2.1 Natural Language Processing and Large Language Models

Natural Language Processing (NLP) enables computers to interpret, analyse, and generate human language. It integrates techniques from linguistics, computer science, and statistics to extract structured information from unstructured text, including clinical notes, referral letters, and patient communications.

Modern NLP relies on machine learning, where models learn patterns from data rather than hand-coded rules. Early NLP systems used simple keyword matching or rule-based grammars, but these fundamentally struggled with the complexity and variability of real-world language (Manning, 2015). The introduction of statistical machine learning methods, such as Support Vector Machines and Naive Bayes classifiers, allowed systems to learn from annotated examples and generalise to unseen text. More recently, deep learning architectures, particularly neural networks with multiple representation layers, have dramatically improved NLP performance across a wide range of analytical tasks (Le Glaz et al., 2021).

The emergence of Large Language Models (LLMs) represents a major advancement in Natural Language Processing. These models are based on transformer architectures, such as BERT (Bidirectional Encoder Representations from Transformers), which utilise self-attention mechanisms to capture contextual relationships between words across extended text sequences (Devlin, Chang, Lee, and Toutanova, 2019). Pre-trained on large corpora, LLMs can be fine-tuned on domain-specific data, including clinical records, to perform tasks such as entity recognition, sentiment analysis, and document classification with high accuracy (Zhang, Schoene, Ji, and Ananiadou, 2022). This enables specialised models for healthcare applications, where precise interpretation of unstructured text is critical.

Task-specific fine-tuned models, such as BERT, differ from general-purpose generative LLMs. Fine-tuned architectures are designed to classify or extract specific information, making them reliable for targeted clinical applications. In contrast, generative LLMs produce novel text and introduce risks of hallucination, reduced transparency, and complex data governance challenges, meaning clinical deployment currently heavily favours task-specific transformers (Haltaufderheide and Ranisch, 2024).

In healthcare, this LLM capability is particularly useful for processing the large volumes of unstructured clinical notes generated daily, enabling the automated extraction of diagnoses,

symptoms, medications, and risk indicators that would otherwise require immense manual triage resources (Ahmed, Kornblum, Oliver, Fusar-Poli, and Patel, 2023). Applying these models provides a consistent mechanism to address the main focus of this research: rapidly identifying subtle indicators of self-harm in referral texts to guarantee high-risk patients are safely prioritised during NHS mental health triage.

2.2 Deep Learning and NLP in Healthcare Decision Support

Deep Learning and NLP are now used in healthcare settings where decisions depend on large volumes of unstructured text. The ability to extract structured meaning from clinical notes, referral letters, and patient records has progressed from theoretical research to operational deployment, particularly in health systems under sustained resource pressure.

The NHS is actively investigating these methods, reflecting a broader shift towards NLP-based decision support in mental healthcare. Reviews show that NLP-driven interventions improve consistency and efficiency by reducing cognitive workload in triage teams (Ni and Jia, 2025). Applied initiatives demonstrate practical viability. For example, the ChronoSig project showed that fine-tuning BERT-based transformers on clinical text can accelerate secondary mental health prioritisation (Health Innovation Oxford, 2024).

Multi-hospital studies validate transformer-based architectures for detecting self-harm risk across diverse clinical settings, demonstrating their ability to learn and isolate linguistic indicators of suicidal ideation (Bey R et al., 2023). These results show that pre-trained language models can be adapted to clinical domains through fine-tuning for explicit risk indicators, including expressions of suicidal ideation and prior self-harm.

Clinical triage depends on subjective interpretation of referral text, which is vulnerable to fatigue and time pressure. Automated systems provide a consistent first-pass mechanism for highlighting risk signals. However, limitations remain. Models trained on one trust's data may not generalise to another, and clinical text varies in structure, completeness, and language style (Ahmed et al., 2023). Automation bias remains a risk, where clinicians over-rely on model outputs at the expense of clinical judgement.

Despite these constraints, evidence suggests that Deep Learning and NLP can substantially enhance healthcare decision support when deployed with appropriate safeguards. The focus must remain on augmenting rather than replacing clinical decision-making, ensuring explainability and preserving clinician accountability.

2.2.1 Evolution of Clinical Natural Language Processing Applications

Clinical NLP has evolved from rule-based systems to deep learning and transformer architectures operating at scale. Early approaches relied on handcrafted rules and dictionaries to extract diagnoses, medications, and symptoms from text. This was followed by traditional machine learning methods, with systematic reviews reporting moderate to strong performance for models such as Support Vector Machines in mental health-related tasks (Le Glaz et al., 2021).

CNNs and LSTMs learned distributed representations directly from data and were applied to sequential electronic health records for detecting suicidal ideation and self-harm risk, typically outperforming classical baselines in F1-score and AUC (Ji, Yu, Fung, Pan, and Long, 2018).

The emergence of transformers such as BERT marked a paradigm shift. Their self-attention mechanisms support modelling of long-range dependencies in complex clinical narratives. Fine-tuned transformers have consistently outperformed earlier approaches across healthcare, legal, and financial document analysis tasks (Pai et al., 2023; Li, 2023; Crema, Attardi, Sartiano, and Redolfi, 2022).

Concurrently, clinical NLP toolkits have matured. Systems such as MedCAT enable concept extraction and ontology linking at scale, supporting population-level analysis of electronic health records. Within the NHS, such pipelines have been applied to mental health monitoring and crisis prediction (Ahmed et al., 2023).

More recently, research has explored large generative models in clinical contexts. However, adoption in mental health decision support remains cautious due to risks including hallucinated outputs, limited transparency, and strict governance constraints. Current NHS-oriented systems therefore prioritise task-specific fine-tuned models with controlled evaluation frameworks rather than general-purpose generative LLMs.

2.2.1.1 Case Study: NLP-Based Triage in NHS Mental Health Services

Recent NHS research has moved beyond classification evaluation toward developing end-to-end triage pipelines that integrate NLP models into referral workflows. One example is the Chronosig project, which developed a transformer-based system to prioritise referrals to secondary mental health services using free-text clinical information (Taylor, Kormilitzin, Lorge, Nevado-Holgado, and Joyce, 2024). The pipeline combines automated deep-learning analysis with clinician oversight, reflecting practical and ethical human-in-the-loop requirements of NHS triage.

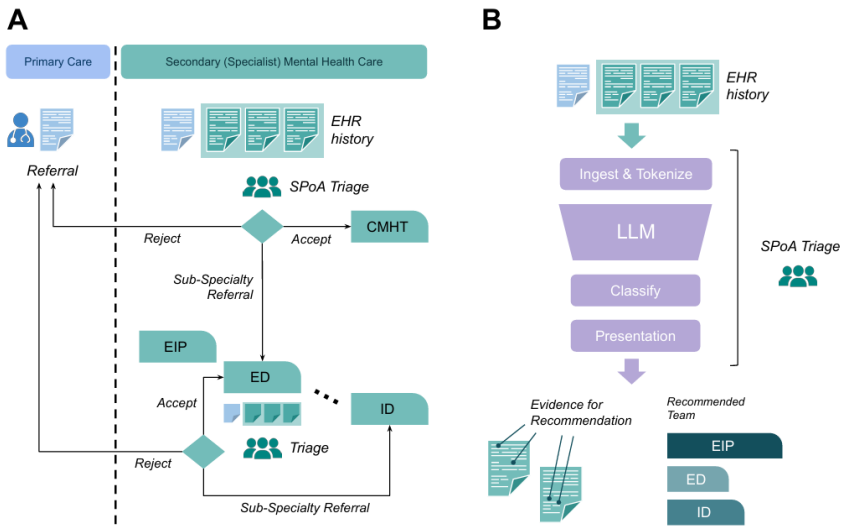

Figure 2.1: End-to-end NLP-driven referral triage pipeline used in NHS mental health services, adapted from the Chronosig project (Taylor, Kormilitzin, Lorge, Nevado-Holgado, and Joyce, 2024).

As shown in Figure 2.1, referral text and historical clinical information are processed using a fine-tuned transformer model to generate a prioritisation score, which is then reviewed by clinicians rather than acted upon autonomously.

This case study demonstrates the transition of NLP models from experimental classification tasks to deployment within realistic clinical workflows. Unlike many research prototypes, the Chronosig pipeline explicitly incorporates human-in-the-loop validation, addressing concerns around clinician trust and patient safety. However, limitations persist. Model performance remains strongly influenced by the quality and representativeness of training data, and systems developed within specific NHS trusts may not generalise across regions with differing documentation practices. Further, the need for clinician oversight emphasises that such systems

currently function as decision-support tools rather than autonomous solutions, reinforcing the importance of maintaining human judgement in high-risk triage scenarios.

2.2.2 Existing Applications and Benchmarks

Performance benchmarks can be contextualised using broader implementations. NLP has found extensive scaling across industries, often demonstrating computational capabilities that parallel or exceed those in mental health. Table 2.1 summarises these benchmarks.

Source	**Domain**	**Methods**	**Metrics**	**Performance**
Ismail et al. (2024)	Customer Service	BERT, LSTM (Sentiment Analysis)	Acc, Prec, Rec, F1	**85–95%** Acc
Grossman and Cormack (2011)	Legal (e-Discovery)	TAR (SVM-based)	Rec, Prec, Elusion	>**75%** Rec, <**5%** Rev
Pai et al. (2023)	Legal (e-Discovery)	FLAN-T5 (LLM)	Prec, Rec, F1	**0.79** F1
Li (2023)	Financial Fraud	CNN, LSTM, BERT	Acc, Prec, Rec, F1	**83.33%** Acc
Le Glaz et al. (2021)	Mental Healthcare	SVM, CNN, LSTM, BERT (review)	Acc, AUC, Prec, Rec	**0.70–0.95+** Acc

Table 2.1: Comparative Analysis of NLP Applications Across Domains

In customer service, transformer-based architectures such as BERT capture complex linguistic patterns to classify customer feedback with state-of-the-art accuracy (Ismail et al., 2024). The legal profession heavily relies on Technology-Assisted Review (TAR) to sift through immense document repositories, with research confirming that TAR is both more efficient and effective than exhaustive manual review (Grossman and Cormack, 2011). Recent studies have fine-tuned open-source LLMs to improve performance in predictive coding tasks (Pai et al., 2023). Financial fraud detection applies deep NLP to parse unstructured text from dense financial reports, with BERT models continuously achieving over 83% accuracy (Li, 2023). These systems utilise a human-in-the-loop workflow, which directly parallels the proposed architectural model for clinical triage.

2.2.2.1 AI-Supported Referral Prioritisation in Clinical Practice

Beyond mental health, similar NLP-driven prioritisation pipelines have been architecturally implemented in other areas of clinical triage. (Abdel-Hafez et al., 2023) propose an AI-assisted referral workflow based on Clinical Prioritisation Criteria (CPC), where machine learning models classify referral urgency from unstructured text. To achieve this, the authors utilised tools such as Amazon Comprehend Medical to extract and standardise medical terminology from referrals, ensuring strict mathematical similarity matching against triage criteria. Although applied outside mental health, this deep learning architecture closely mirrors NHS triage processes.

Figure 2.2 illustrates the clinical referral pathway, identifying the third phase (specialist triage) as the exact juncture where machine learning models are embedded to assist clinicians. By injecting the predictive architecture directly into this specific step rather than operating it

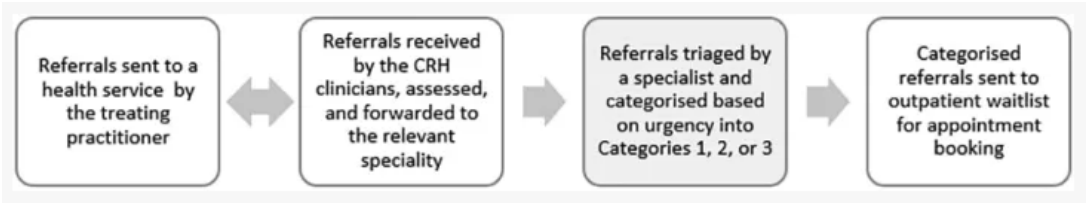

Figure 2.2: AI-assisted referral prioritisation workflow using Clinical Prioritisation Criteria (CPC), adapted from (Abdel-Hafez et al., 2023).

as a standalone oracle, this human-in-the-loop design aligns seamlessly with NHS governance requirements and mirrors the architectural principles necessary for mental health triage

2.2.3 Limitations of Existing NLP-Based Triage Systems

Despite demonstrating feasibility, a broader examination of existing systems such as ChronoSig and CPC reveals several systematic limitations related to performance robustness, predictive calibration, and live clinical applicability.

Reported evaluations of Chronosig indicate strong aggregate performance on retrospective referral datasets; however, external validation across different NHS trusts and diagnostic presentations is still ongoing. Performance is expected to vary across risk categories and clinical presentations, particularly for patients with less explicit or indirectly expressed indicators of abstract distress, as is common in atypical presentations and certain demographic groups (Joyce et al., 2021). This indicates reliance on overt linguistic markers, limiting effectiveness in cases where genuine risk is conveyed implicitly or through nuanced temporal context.

Similarly, CPC-based models achieve competitive baseline classification metrics (Abdel-Hafez et al., 2023), but are constrained by annotation uncertainty. Risk labels are historically derived from clinician judgement rather than objective mathematical outcomes, directly introducing label noise that naturally affects achievable target accuracy (Sylolypavan, Sleeman, Wu, and Sim, 2023). Inter-rater variability in human mental health assessment further complicates performance interpretation, as computational misclassifications may actually reflect ambiguity in human ground-truth rather than genuine NLP model error.

Both systems prioritise recall parameters to minimise false negatives, a design choice strongly aligned with patient safety objectives. However, their published results indicate severely elevated false positive rates, which reduces pipeline practical utility by actively increasing clinician workload and diminishing trust in automated triage probability recommendations (Brann, Sterling, Frisch, and Schrager, 2024). Such trade-offs directly highlight limitations in precision that are not fully captured by reviewing aggregate F1 or AUC metrics in isolation.

Cross-domain generalisation remains a critical challenge. Existing neural benchmarks are typically trained and evaluated on datasets sourced from a limited number of clinical contexts. Prior systemic studies caution that transformer-based models may falsely overfit stylistic or institutional artefacts embedded within local data, rapidly resulting in degraded predictive calibration when deployed to unseen healthcare settings or diverse populations (Cohen et al., 2022).

Current evaluations rely on offline performance metrics such as precision, recall, and AUC. While highly suitable for isolated mathematical benchmarking, these offline measures do not naturally capture downstream clinical impact, user interaction effects, or workflow decision-support effectiveness in live real-world settings. Prior leading studies in clinical Machine Learning strictly caution that strong offline deep learning performance does not automatically translate into noticeably improved decision-making or workflow integration, leaving open questions regarding

long-term operational reliability and practical accuracy (Kelly, Karthikesalingam, Suleyman, Corrado, and King, 2019; Topol, 2019).

These limitations indicate that while existing NLP-based triage systems establish a highly robust technological foundation, their reported performance and accuracy remain constrained by input data quality, human annotation subjectivity, and limited algorithmic generalisability. Directly addressing these specific computational gaps, particularly through exploring probability calibration boundaries and constructing a clinician-in-the-loop Active Learning pipeline, forms the overall primary methodological motivation of this dissertation project.

2.3 Self-Harm Risk Detection in Mental Health

Detecting self-harm risk from clinical text is a safety-critical NLP application. Systemic patterns of language in referral letters and clinical notes can be computed to reliably predict patients at elevated risk (Bey R et al., 2023; Cohen et al., 2022). These approaches typically combine granular entity recognition, such as identifying tokenised mentions of prior self-harm, abstract suicidal ideation, or explicitly named psychiatric medications, with broad document-level classification to assign a holistic risk probability score (Ji et al., 2018).

A major ongoing technical architecture impediment in this classification domain focuses on distinguishing between active, immediate suicidal ideation versus historical, managed self-harm, as well as correct handling clinical negation (for example, "patient denies current thoughts of self-harm"). While traditional keyword-matching systems often failed on these linguistic nuances, modern bidirectional transformer architectures fundamentally excel at implicitly capturing the contextual and temporal indicators of risk. Research by Cohen et al., 2022 highlights the immense complexities of successfully predicting suicide risk from open-ended verbal interview language captured in emergency departments, noting that vulnerable patients frequently convey critical distress through implicit or highly atypical language structures rather than via explicitly programmed high-risk alert keywords.

State-of-the-art NLP benchmarks strongly demonstrate the efficacy of these advanced models across highly diverse textual domains. Bey R et al., 2023 successfully validated applied NLP models for mental health risk detection simultaneously across multiple hospital architectures, mathematically proving that deep learning can isolate and extract generalisable risk tensor signals from highly complex, noisy electronic health records. Similarly, supervised algorithmic learning techniques have actively achieved strong performance in identifying suicidal ideation within notably less structured online internet user content (Ji et al., 2018). However, substantially translating these high theoretical classification accuracy and AUC scores into genuinely operational clinical decision tools requires highly rigorous probability calibration frameworks.

The key challenge is balancing sensitivity (catching all true high-risk cases) alongside statistical specificity (minimising resource-wasting false alarms). In high-pressure live clinical contexts, false negatives (missing a high-risk patient in crisis) carry incredibly severe safety consequences, whereas excessive false positives consume scarce human triage resources. Technical model development and loss-functions must therefore directly prioritise recall parameters, often operating at the deliberate expense of raw precision, to ensure strict patient safety (Le Glaz et al., 2021). Thus, to remain operationally safe and viable, predictions must be presented to clinicians with uncertainty estimates and XAI evidence highlighting linguistic features contributing to risk scores.

2.4 Technical Benchmarks and Evaluation

Evaluation uses standard train/validation/test splits (80/10/10 ratios) and rigid k-fold cross-validation loops. The primary target metrics are Precision, Recall, F1-Score, and AUC-ROC. Within specifically healthcare software applications, sensitivity and specificity parameters are also important, as they directly correlate to genuine clinical utility and govern the balance between dangerously missed high-risk cases versus burdensome unnecessary triage alarms (Le Glaz et al., 2021).

Pre-trained baseline transformer architectures, such as BERT and RoBERTa, are commonly fine-tuned on targeted domain-specific text pipeline data using established deep learning tensor frameworks like PyTorch together with the highly established Hugging Face Transformers Python library (Wolf et al., 2020; Hugging Face, 2024a). These models require careful hyperparameter tuning (learning rate, gradient batch sizing, and absolute number of training epochs) alongside automated early stopping to prevent catastrophic data overfitting. Specifically within modern mental health research, fine-tuned bidirectional BERT models have been consistently used to successfully classify unstructured online internet posts or heavily formatted clinical medical text into distinct suicidal versus non-suicidal classification categories, achieving remarkably strong statistical performance across all standard metrics benchmarks (Ji et al., 2018).

2.4.1 Implementation with PyTorch and Transformers

This subsection outlines a standard fine-tuning pipeline for BERT-style models, providing a clear reference for the implementation choices discussed in Section 3 rather than proposing a new method.

Fine-tuning a pre-trained BERT model for mental health risk classification follows a standardised tensor-based workflow in PyTorch, as demonstrated in Hugging Face Transformers and PyTorch documentation (Hugging Face, 2024b; PyTorch, 2024). The process begins by loading a pre-trained model and its associated tokenizer:

```

from transformers import BertTokenizer,
    BertForSequenceClassification

model_name = 'bert-base-uncased'
tokenizer = BertTokenizer.from_pretrained(model_name)
model = BertForSequenceClassification.from_pretrained(
    model_name,
    num_labels=2  # high risk vs not high risk
)
```

Listing 2.1: Loading a Pre-trained BERT Model

In listing 2.1 the neural model is not trained from scratch. Instead, a general-purpose BERT model is loaded into active memory and adapted for a targeted simple two-class problem (high risk versus not high risk). Fine-tuning modifies the geometric tensor weights using the referral text corpus, aligning classification boundaries with the specific nuances of mental health triage diagnostics.

Input clinical referral text data is then tokenised into structured input algorithmic IDs and positional attention memory masks, formatting the strings to become strictly suitable for internal processing within deep transformer mathematical models:

```

```

```
text = "Patient reports persistent suicidal ideation and recent self-harm."

inputs = tokenizer(
    text,
    padding=True,
    truncation=True,
    max_length=512,
    return_tensors='pt'
)

# inputs['input_ids'] and inputs['attention_mask'] are passed to the model
```

Listing 2.2: Tokenising Referral Text

Listing 2.2 showcases the raw natural human English sentence string is converted into discrete, rigid numerical tensor tokens and an accompanying geometirc attention mask so that the abstract semanitc sequence can be linearly processed by the internal transformer model layers. In practice, the same procedure is applied to every referral in the dataset, producing consistent input representations for training and evaluation.

Training uses minibatches of tokenised examples and an optimiser such as AdamW, following the pattern in the Hugging Face text classification examples (Hugging Face, 2024b):

```
from torch.utils.data import DataLoader
from transformers import AdamW

train_loader = DataLoader(train_dataset, batch_size=16, shuffle=True)
optimizer = AdamW(model.parameters(), lr=2e-5)

model.train()
for epoch in range(3):
    for batch in train_loader:
        input_ids = batch['input_ids']
        attention_mask = batch['attention_mask']
        labels = batch['labels']

        outputs = model(
            input_ids=input_ids,
            attention_mask=attention_mask,
            labels=labels
        )
        loss = outputs.loss

        optimizer.zero_grad()
        loss.backward()
        optimizer.step()
```

Listing 2.3: Simplified Fine-tuning Loop

Listing 2.3's loop repeatedly shows batches of referral examples to the model, compares its predictions with the true labels, and nudges the model parameters to reduce the loss. Hyperparameters such as batch size, learning rate, and number of epochs control how quickly the model learns and how likely it is to overfit the training data.

After training, the model is evaluated on a held-out test set. Predictions are converted to labels and standard metrics are computed using libraries such as scikit-learn, as is common in published work on suicidal ideation detection (Ji et al., 2018):

```python


from sklearn.metrics import classification_report, roc_auc_score

model.eval()
predictions, true_labels = [], []

with torch.no_grad():
    for batch in test_loader:
        input_ids = batch['input_ids']
        attention_mask = batch['attention_mask']
        labels = batch['labels']

        outputs = model(input_ids=input_ids,
                        attention_mask=attention_mask)
        logits = outputs.logits
        preds = logits.argmax(dim=1)

        predictions.extend(preds.cpu().numpy())
        true_labels.extend(labels.cpu().numpy())

print(classification_report(true_labels, predictions,
      target_names=['Not High Risk', 'High Risk']))
auc = roc_auc_score(true_labels, predictions)
print(f"AUC-ROC: {auc:.3f}")
```

Listing 2.4: Evaluation and Metric Calculation

Once training is complete, the model is tested on unseen data. The code computes standard classification metrics such as precision, recall, F1-score, and AUC-ROC, which provide an objective summary of how well the model distinguishes high-risk from lower-risk referrals.

This workflow closely mirrors the implementations described in mental health NLP studies that fine-tune BERT for risk classification tasks, where reported AUC and F1-Score values typically exceed those of traditional machine learning baselines (Ji et al., 2018).

2.5 Integration into NHS Workflows and Ethical Governance

Beyond model performance, successful integration of these NLP algorithms into NHS triage also depends on organisational, ethical, and governance factors. Technical accuracy alone is not sufficient; stakeholder buy-in, robust data governance, and mitigation of known risks are equally important. Clinicians must understand how the algorithm generates scores, and the system must be designed as decision support (highlighting candidates for urgent review) rather than decision replacement (automatically determining triage priority).

Data governance is fundamental. All patient data used for training, validation, and testing must be anonymised, stored securely, and accessed only by authorised personnel. Models must be regularly audited for bias, particularly across patient demographics, and performance must be continuously monitored in live settings to detect model drift or unintended failures.

Ethical considerations include the responsibility to avoid reinforcing historical biases in data, ensuring that false positives and false negatives are clearly communicated to clinicians, and maintaining transparency regarding model limitations.
These considerations build on the ethical themes outlined in Chapter 1 and frame the constraints within which any Deep Learning NLP-based triage prototype must operate in NHS mental health services

2.6 Literature Synthesis

Transformer models outperform traditional NLP in mental health risk detection but remain constrained by poor generalisation and subjective annotations. Prioritising recall reduces false negatives at the cost of increased clinician workload, positioning these systems as decision-support tools rather than autonomous triage solutions.

Source	**Sample & Context**	**Methodology**	**Key Findings**	**Limitations**
Taylor et al., 2024	NHS secondary mental healthcare waiting lists (ChronoSig)	Bespoke LLMs / transformers on unstructured EHR notes	Produces triage recommendations consistent with clinical practice, with potential to improve referral processing efficiency	Single-GPU constraints; reliance on highly curated local trust data
Abdel-Hafez et al., 2023	17,378 specialist outpatient referrals evaluated against CPC	Supervised ML categorising clinical prioritisation criteria	Assists clinicians in referral classification, improving triage efficiency	Moderate agreement (53.8%), reflecting variability and subjectivity in clinical annotation
Cohen et al., 2022	Open-ended patient interviews from Emergency Departments	NLP applied to unstructured interview transcripts	Demonstrates effective discrimination of suicide risk from linguistic features	Limited by small sample size and controlled evaluation setting.
Bey R et al., 2023	Multi-hospital EHR data (2.9M records)	NLP-based extraction of suicidality indicators from clinical text	Enables large-scale monitoring of suicide attempts and risk factors	Limited by retrospective design and institutional data variability
Haltaufderheide and Ranisch, 2024	Systematic review of 796 healthcare AI studies	Analysis of generative LLM applications in healthcare	Highlights benefits in data analysis and decision support	Concerns around hallucination, bias, transparency, and privacy
Le Glaz et al., 2021	Systematic review across mental health disciplines	Comparison of deep learning and traditional NLP methods	Demonstrates strong performance in retrospective mental health tasks	Limited translation to real-world clinical deployment

Source	Sample & Context	Methodology	Key Findings	Limitations
Sylolypavan et al., 2023	ICU datasets annotated by 11 clinical experts	Supervised learning under annotation inconsistency	Demonstrates variability in model outputs due to expert disagreement	Lack of consistent ground truth limits model reliability and validation
Klug et al., 2024	Clinical NLP across the hospital patient journey	Systematic review of NLP applications in healthcare workflows	Identifies opportunities for NLP in clinical decision support across patient stages	Limited real-world deployment; uneven research coverage across clinical processes

Table 2.2: Synthesis of Key Literature in NLP-Driven Clinical Triage and Mental Health Risk Detection.

Chapter 3

Methodology

This study adopts a controlled experimental methodology to evaluate a proof-of-concept transformer-based NLP triage system. Given that the model was trained on synthetic data and not integrated into live NHS infrastructure, the evaluation focuses on methodological feasibility, model behaviour, and system-level performance rather than clinical deployment.

The research design separates model development from downstream clinical evaluation, enabling classification performance to be assessed independently from prioritisation behaviour. This distinction is critical for analysing whether improvements in predictive accuracy translate into effective triage ordering within a decision-support context.

Figure 3.1 presents the end-to-end methodological workflow of the proposed system, spanning data generation, model development, and evaluation within a simulated NHS triage environment. The pipeline explicitly distinguishes classification outputs from downstream prioritisation processes, allowing the study to identify any misalignment between predictive performance and clinical utility.

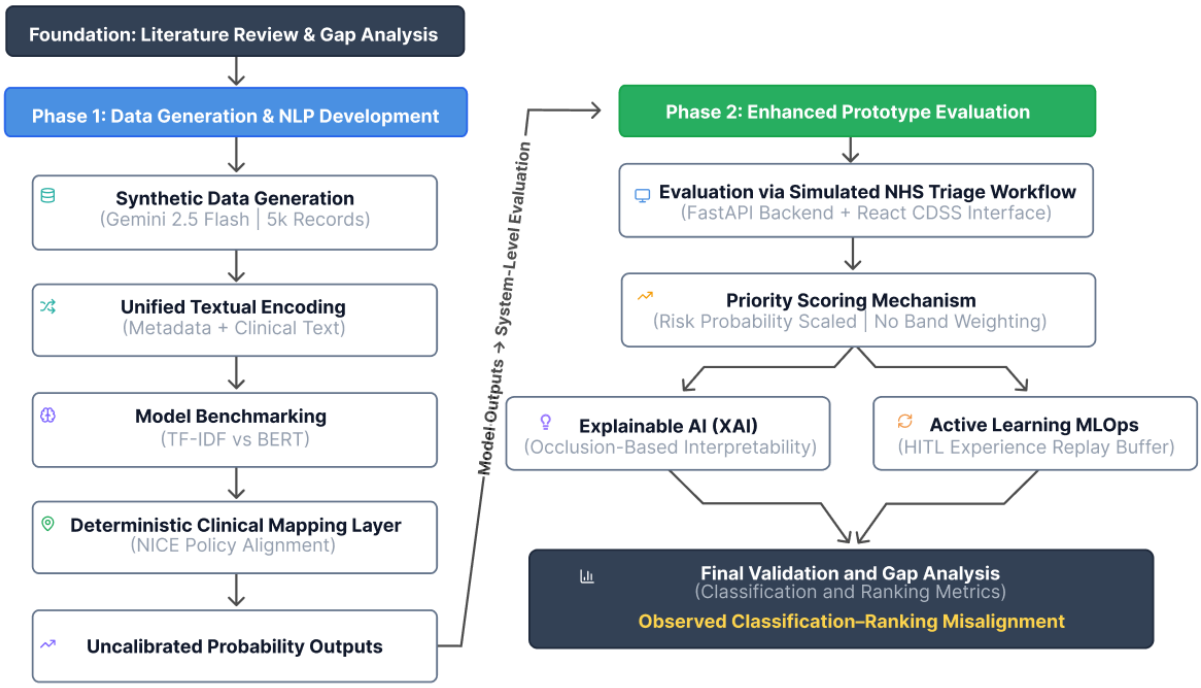

Figure 3.1: Methodological workflow of the proposed NLP-based triage system.

3.1 Data Generation and NLP Development

To satisfy Objective 2, the methodology first establishes an automated synthetic data pipeline for generating clinical referral text, before transitioning from traditional machine learning (ML) classification approaches to transformer-based architectures for self-harm risk detection. This design enables a controlled comparison between a traditional sparse-feature baseline (TF-IDF with Logistic Regression) and a contextual transformer-based model (BERT), allowing observed performance differences to be attributed to model architecture rather than data variation.

3.1.1 Synthetic Dataset Generation

Given strict NHS data governance and GDPR regulations, obtaining real-world patient psychiatric records within an undergraduate scope is not feasible within project constraints. To solve this, the methodology employed an automated synthetic data pipeline (backend/nlp/data_augmenter.py). Using the Gemini 2.5 Flash foundation LLM guided by strict clinical prompt constraints, 5,000 synthetic patient records were generated, encompassing demographics, symptoms, and classified risk labels (Low, Medium, High). This enabled an ethically compliant and scalable dataset, consistent with recent work leveraging LLM-generated text for suicide-related NLP tasks (Ghanadian, Nejadgholi, and Al Osman, 2024).
Synthetic data may not fully capture the linguistic variability, ambiguity, and noise present in real clinical text. As such, the generated dataset was treated as an approximation of referral-style input, and the impact of this limitation was explicitly considered during evaluation.

3.1.2 Algorithm Selection and Enhancement

The research established a baseline algorithm using TF-IDF text vectorisation coupled with a Logistic Regression classifier to map multi-class risk labels to triage priority bands. To address the limitations of sparse feature representations, the methodology extended the baseline by introducing a pre-trained HuggingFace BERT transformer as the core encoder model. BERT was selected for its bidirectional contextual representations, enabling the model to capture dependencies across referral text that are not accessible to sparse feature methods. This is critical for detecting implicit self-harm indicators, where meaning is distributed across phrases rather than isolated keywords, enabling more reliable detection of implicit self-harm indicators and nuanced semantic patterns in referral text (Crema et al., 2022). The end-to-end inference pipeline, including probability estimation, priority assignment, and Explainable AI attribution, is implemented as shown in Listing D.2.

3.1.3 Unified Textual Encoding and Calibration

A key architectural consideration in this development pipeline was how the model handled heterogeneous data types. Rather than constructing complex multi-branch architectures to process structured and unstructured data separately, this study adopted a unified textual encoding approach. Structured metadata (e.g., age and gender) was embedded directly into the referral-style text prior to tokenisation, enabling the transformer model to process all inputs within a single sequential representation. Tokenisation and tensor encoding are implemented as shown in Listing D.3. This design simplifies the pipeline while preserving the contextual signals required for accurate risk detection.

Following this, the model applied a linear classification head and softmax function to produce probabilistic outputs. These raw probabilities were mapped into priority bands using fixed decision thresholds. Priority band assignment and queue prioritisation logic are implemented as shown in Listing D.1.

Although calibration is typically required to ensure that predicted confidence values can be interpreted as reliable risk likelihoods, it was intentionally omitted in this study to isolate the transformer's unadjusted probability distributions.

The prioritisation score was derived directly from the predicted class probability, scaled to a 0–100 range, enabling ranked queue ordering while preserving intra-class confidence differences.

3.1.4 Post-Processing Clinical Guidance Mapping

As a post-processing constraint to support NHS governance requirements, the methodology incorporated a deterministic post-processing mapping layer. Priority bands were used as keys to retrieve predefined NICE-aligned triage protocols (National Institute for Health and Care Excellence (NICE), 2022). This layer appends fixed clinical guidance to the model output without influencing the underlying prediction, ensuring consistency while eliminating generative hallucination risk.

3.2 Evaluating the Enhanced Prototype

To fulfil Objective 3, evaluating the algorithm within a simulated clinical scenario, the methodology prioritised system-level interaction and safety over isolated algorithmic testing.

3.2.1 Simulated NHS Triage Workflow

Deploying an experimental NLP-based decision-support prototype into a live, high-stakes NHS environment poses significant safety risks. Therefore, the chosen methodology constructed a full-stack simulated triage environment, consisting of a FastAPI backend and a React-based frontend dashboard presented in figures A.1, A.2 respectively. This approach balanced safety with ecological validity, enabling the system to be evaluated as a Clinical Decision Support System (CDSS). The simulation modelled how clinicians would interact with prioritised outputs within a structured triage workflow, rather than treating the system as an autonomous decision-maker.

3.2.2 Explainable AI (XAI) and Active Learning MLOps

A crucial aspect of the evaluation was clinical trust and model longevity. To prevent the model from acting as an opaque "black box," occlusion-based Explainable AI (XAI) was deployed to highlight the specific textual tokens influencing the algorithm's decisions, supporting interpretability and transparency (figure A.2).

Furthermore, to address the challenge of static model degradation (concept drift), a Human-in-the-Loop (HITL) MLOps Active Learning pipeline was integrated into the workflow. When clinicians confirmed or overrode a suggested priority band during evaluation, this feedback was captured within a programmatic Experience Replay Buffer. This buffer combined new corrections with historical training data to support iterative retraining of the PyTorch model over time. Replay-based training was specifically adopted to alleviate gradient interference by interleaving historical and newly acquired samples during optimisation. This approximates a stable training distribution and successfully mitigates catastrophic forgetting (Rolnick, Ahuja, Schwarz, Lillicrap, and Wayne, 2019).

3.2.3 Evaluation Metrics and Assumptions

Model performance was evaluated by comparing the baseline Logistic Regression and enhanced BERT models using standard classification metrics (Accuracy, Precision, Recall, F1-Score, AUC-ROC), with formal definitions provided in Appendix E. These metrics reflect the clinical trade-off between false negatives and false positives, where recall is critical for patient safety and precision impacts clinician workload.

The primary limitation remained the use of synthetic data, which may not capture the full complexity of real clinical language; therefore, results indicate methodological feasibility rather than readiness for clinical deployment.

This evaluation therefore reflects internal comparative performance under controlled conditions rather than externally validated clinical effectiveness.

Chapter 4

System Architecture and Design

4.1 Overview

The system architecture extends prior NLP-driven triage pipelines, including the ChronoSig framework (Taylor et al., 2024) and AI-assisted Clinical Prioritisation Criteria workflows (Abdel-Hafez et al., 2023), by incorporating calibrated risk scoring, explicit prioritisation logic, and clinician-facing decision support. While these earlier systems demonstrated the feasibility of applying NLP to triage, they did not fully address how classification outputs translate into reliable prioritisation behaviour within NHS mental health settings. This work therefore introduces an architecture that explicitly separates prediction, calibration, and ranking, implemented as a sequential pipeline from text ingestion to prioritised triage queue generation.

A lightweight, web-based clinician-facing dashboard was developed to securely display prioritised referral cases that had been processed by a deep-learning NLP backend model. This predictive model, trained on anonymised synthetic datasets, produced calibrated risk scores and corresponding suggested priority bands. These outputs were then consumed by the frontend interface to order active clinical cases within a clinician-facing triage queue, allowing for safe human-in-the-loop review and feedback.

4.2 High-Level System Architecture

Figure 4.1 illustrates the overall architecture of the LLM-driven NLP prioritisation prototype. Rather than relying on a patient-facing data collection portal, synthetic referral-style text and patient metadata were ingested directly by a FastAPI-based backend (through SwaggerUI) and processed by an NLP risk scoring module powered by a fine-tuned BERT-based transformer model. The resulting risk probability was fed into a prioritisation and triage component that mapped scores to priority bands, generated a continuous prioritisation score, and exposed a sorted referral queue through a clinician triage interface for human-in-the-loop review, rather than automatic action.

This separation between prediction, prioritisation, and post-processing guidance was intentional, allowing the classifier to remain analytically independent from downstream safety and governance mechanisms.

To ensure long-term clinical safety, the system architecture also established a continuous feedback loop; clinician overrides were captured via a regulatory audit ledger to feed an MLOps Active Learning replay buffer, while algorithmic predictions were physically mapped to action protocols using a deterministic clinical protocol mapping module.

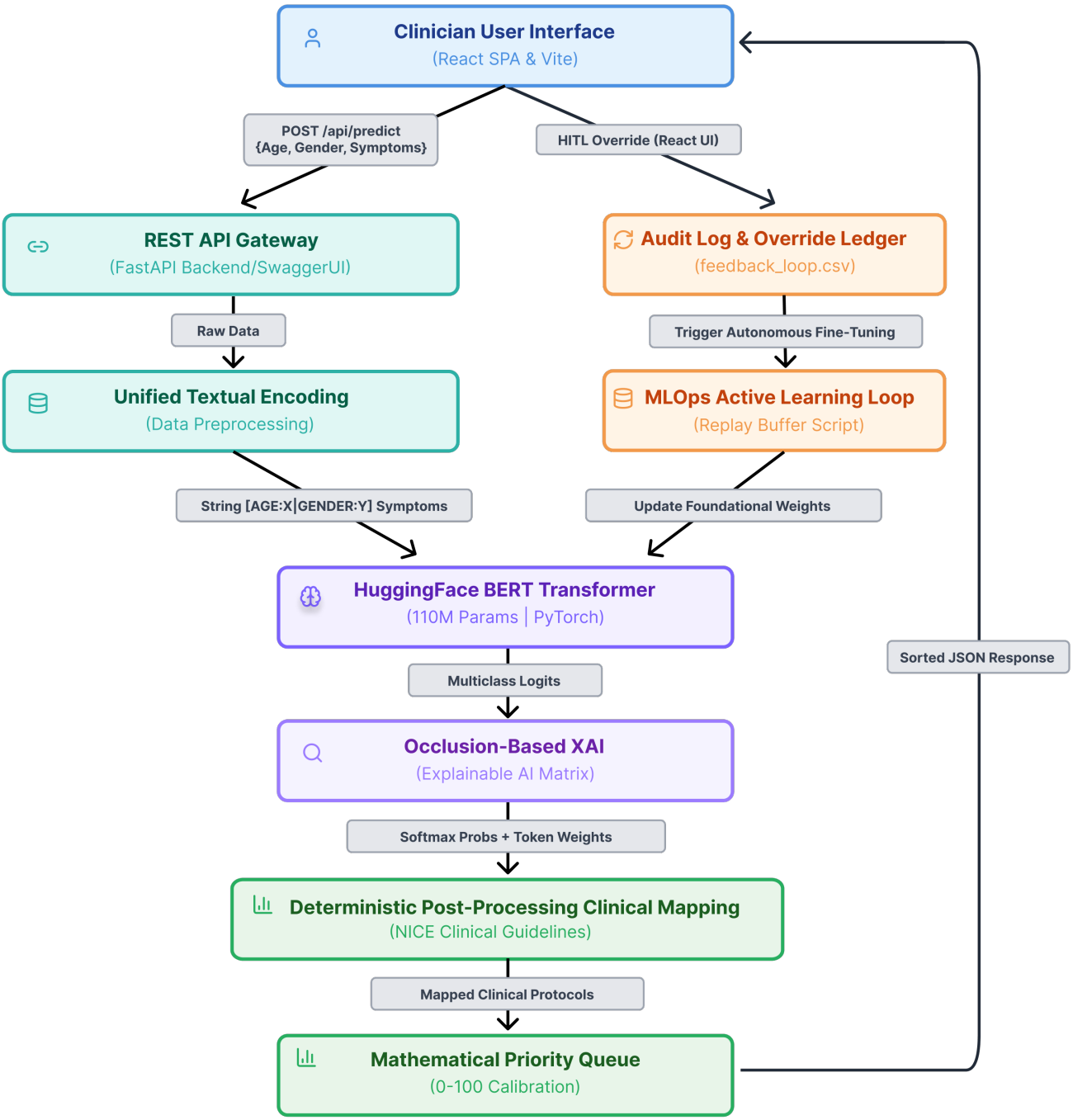

Figure 4.1: High-level architecture of the transformer-based NLP prioritisation prototype used in this dissertation.

4.3 Architectural Logic (Baseline vs Enhanced)

Figure 4.2 compares the baseline and enhanced NLP classification pipelines used in this study. The baseline route applied standard preprocessing and TF–IDF feature extraction followed by a logistic regression classifier that output a binary risk label, which was then mapped to a simple priority category and used to construct a sorted queue. In contrast, the enhanced route employed a BERT encoder and classification head to estimate a continuous risk probability, which was calibrated, thresholded into high, medium, and low priority bands, and then passed into a prioritisation module that produced an ordered

triage list. The prioritisation module therefore operates as a downstream ranking layer, transforming calibrated model outputs into queue-ordering behaviour rather than treating raw class predictions as sufficient for triage sequencing.

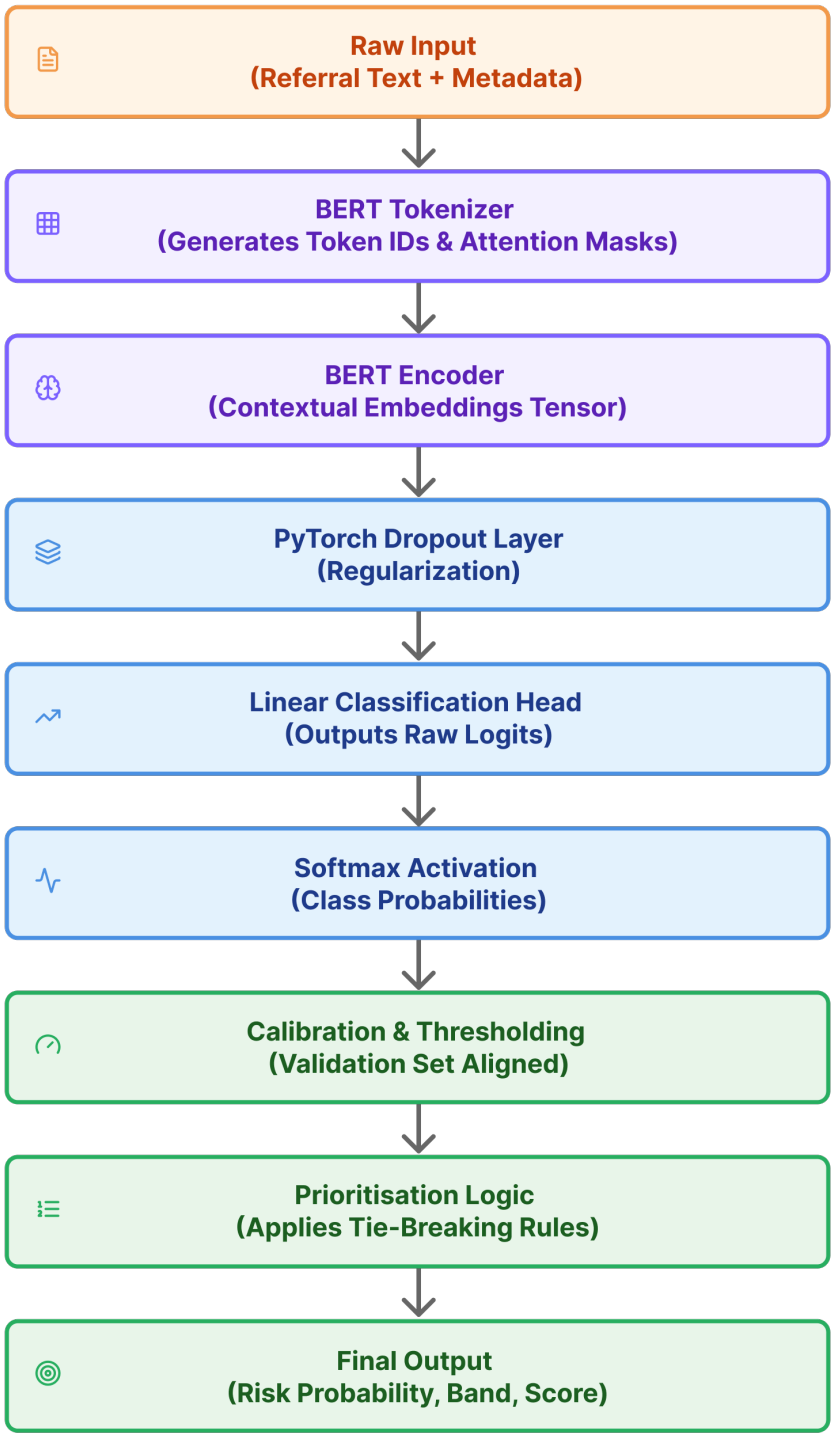

Figure 4.2: Comparison between the baseline TF–IDF plus logistic regression prioritisation pipeline and the enhanced BERT-based LLM-driven prioritisation pipeline.

This architectural separation directly supports the controlled experimental comparison defined in the methodology, ensuring that differences between baseline and enhanced systems arise from model capability rather than downstream prioritisation logic.

4.4 Unified Textual Data Representation

To avoid the architectural complexity of constructing separate processing pipelines for structured and unstructured data, the system adopted a unified textual representation strategy. Structured tabular metadata (e.g., age and gender) was embedded directly into the referral-style text prior to tokenisation. This enabled the transformer model to process all inputs within a single sequential representation, allowing implicit modelling of relationships between demographic attributes and linguistic context through self-attention.

As illustrated in Figure 4.3, this approach removes the need for multi-branch neural architectures while still incorporating heterogeneous information into the classification process in a computationally efficient manner.

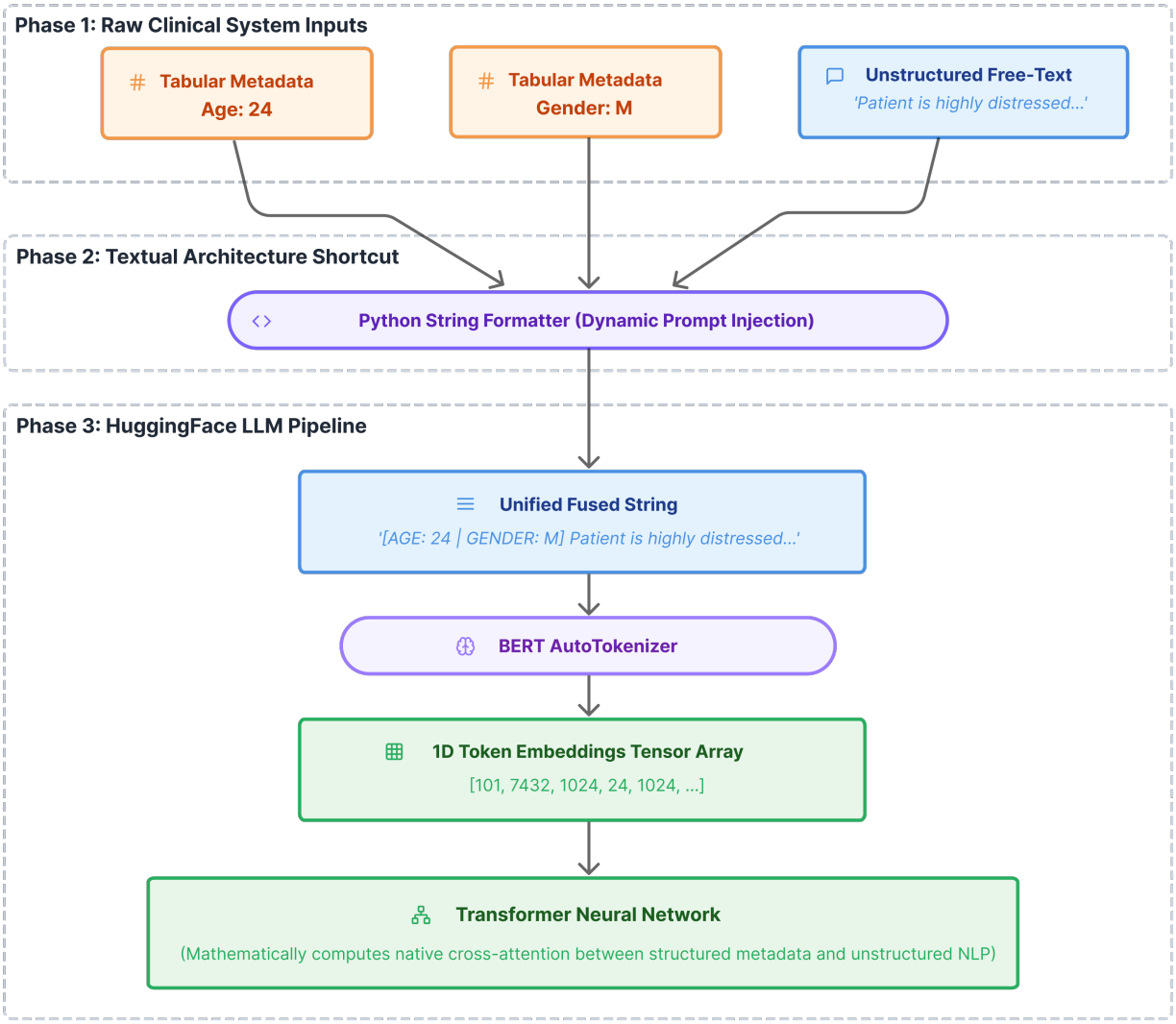

Figure 4.3: Unified textual representation approach, embedding structured metadata directly into referral text prior to tokenisation to enable joint processing within a transformer model.

4.5 Clinician Interface and Simulated Clinical Workflow

To evaluate the system in a way that was compatible with the constraints of undergraduate research and NHS data governance, the prototype operated within a simulated mental health triage workflow, shown in Figure 4.4. On the input side, simulating a primary care submission via the backend API (SwaggerUI), referral-like text was ingested, fused with demographic metadata, processed by the NLP transformer model, and converted into a suggested priority band. The prioritisation module then dynamically updated the sorting metrics and presented a prioritised list of open referrals on the React front-end.

On the clinician side, psychiatric staff accessed this queue through a secure triage interface. They inspected individual referrals alongside the model-generated risk score, priority band, and occlusion-based Explainable AI (XAI) textual highlighting. From here, the clinician could confirm, upgrade, or downgrade the LLM's probabilistic suggestion.

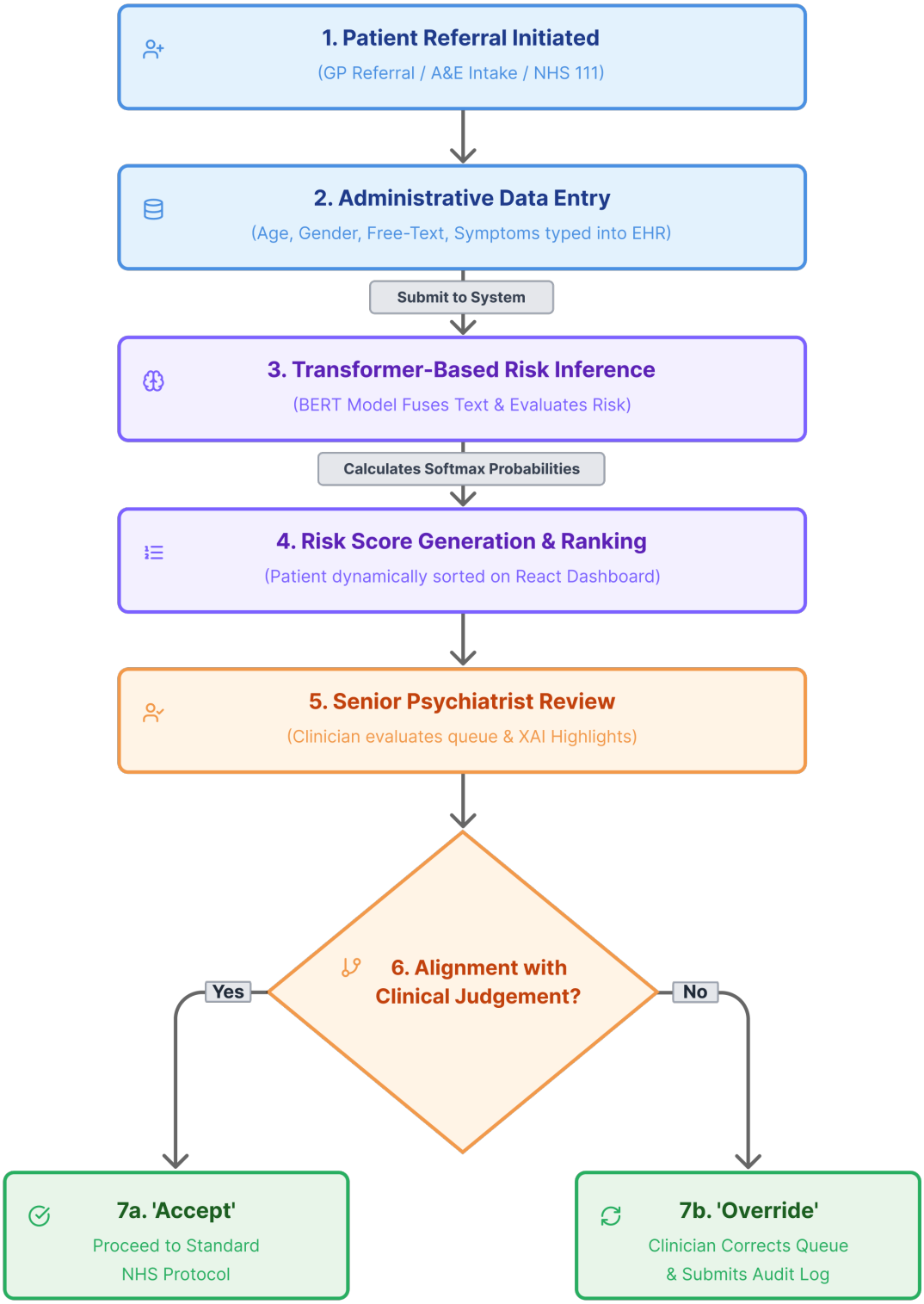

Figure 4.4: Simulated NHS mental health triage workflow illustrating how transformer-based NLP outputs support, rather than replace, clinician prioritisation decisions.

Chapter 5

Implementation

5.1 Synthetic Dataset Generation Workflow

Given the sensitivity of clinical data and the strict regulatory constraints surrounding patient information (e.g., GDPR and NHS data governance policies), access to real-world triage datasets was not feasible within the scope of this study. To address this limitation, a synthetic dataset was generated using an end-to-end programmatic generation pipeline, illustrated in Figure 5.1.

A total of 5,000 synthetic clinical records were programmatically generated using the Gemini 2.5 Flash large language model via a secure Python pipeline (backend/nlp/data_augmenter.py). The generation process was guided by engineered structural prompts designed to produce diverse psychiatric presentations, incorporating demographic metadata (Age, Gender), symptoms, medical history, and multi-class triage risk categories (Low, Medium, High). The full prompt specification is provided in Appendix C.

Class balance constraints were implicitly controlled through prompt engineering to avoid severe label skew, ensuring that model performance comparisons were not artificially influenced by class imbalance.

Listing 5.1 demonstrates the algorithmic constraints established within the Python pipeline to enforce clinical fidelity during API interaction.

```
prompt = f"""
You are an expert UK NHS psychiatric triage nurse. Generate exactly
    {batch_size} synthetic, highly realistic mental health patient
    referral texts.
They must be diverse in age (18-80), gender, and symptomatology.
Ensure a realistic distribution of:
- "High" Priority (acute crisis, active suicide planning).
- "Medium" Priority (severe impairment, urgent but safe).
- "Low" Priority (routine therapy, mild anxiety).
"""
response = client.models.generate_content(
    model='gemini-2.5-flash',
    contents=prompt,
)
```

Listing 5.1: Gemini 2.5 Flash strict prompt constraints used in the data augmentation pipeline.

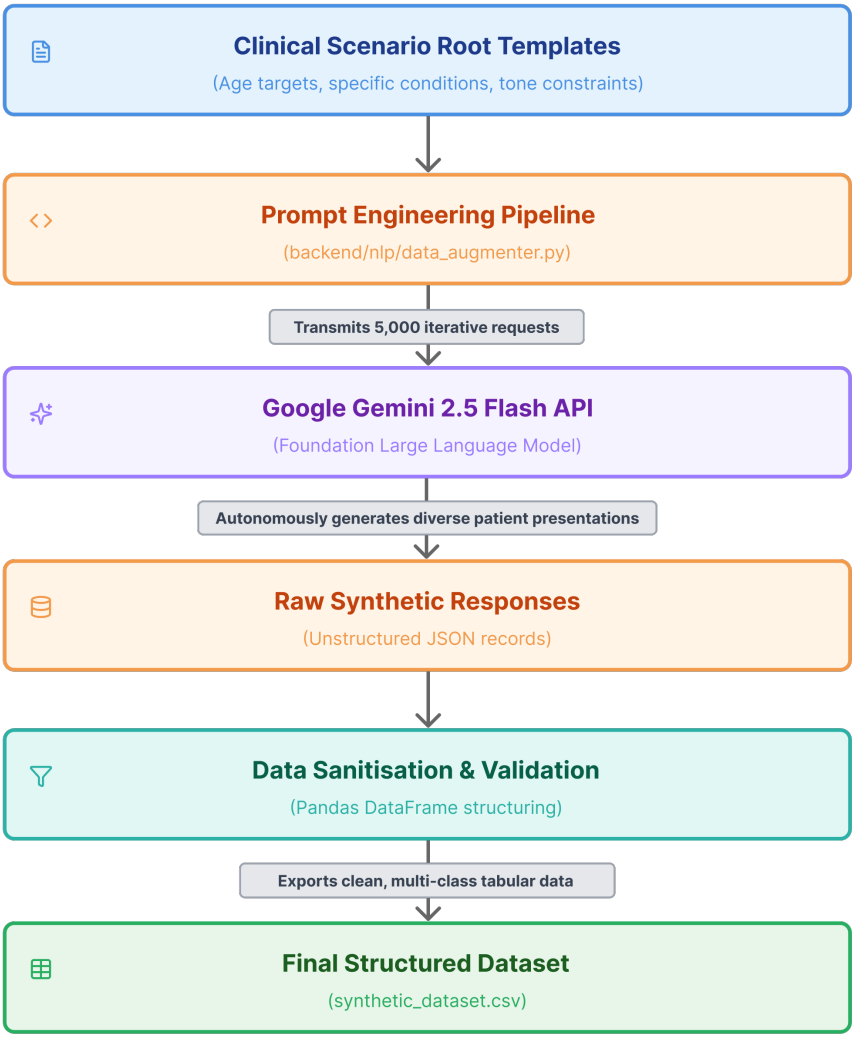

Figure 5.1: Overview of the autonomous synthetic data augmentation pipeline leveraging LLM-based clinical prompt engineering to produce a structured 5,000-record dataset.

All generated records were fully anonymised by design, containing no identifiable patient information.

5.2 Deep Learning Model Development and PyTorch Tensors

The internal structure of the enhanced transformer-based NLP pipeline is illustrated in Figure 5.2. The fused referral text and metadata were transformed into token IDs and attention masks by the BERT tokenizer, encoded into contextual embeddings by the BERT encoder, and passed through dropout and a linear classification layer to obtain logits. Softmax was applied to produce class probability distributions.

Priority bands were assigned using fixed thresholds on raw softmax outputs. Calibration techniques (e.g., temperature scaling or isotonic regression) were intentionally omitted to isolate the transformer's unadjusted probability distributions and assess their suitability for downstream prioritisation.

A prioritisation logic block then converted the predicted probabilities and assigned risk band into a sortable score, applying tie-breaking rules where required, and outputting a risk probability, priority band, and prioritisation score for each referral. The prioritisation score was derived directly from the predicted class probability, scaled to a 0–100 range. Because higher-risk predictions naturally produce higher probabilities, this design ensures that high-risk referrals dominate queue ordering while preserving

intra-band ranking by confidence.

Listing 5.2 highlights the explicit HuggingFace Trainer configurations utilised during this phase, including batch sizes and warmup steps selected.

```
training_args = TrainingArguments(
    output_dir=save_dir,
    num_train_epochs=3,
    per_device_train_batch_size=16,
    warmup_steps=65, # Calculated ~10% for gradient convergence
    weight_decay=0.01,
    eval_strategy="epoch",
    load_best_model_at_end=True,
)
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=val_dataset,
    compute_metrics=compute_metrics
)
```

Listing 5.2: Training hyperparameters optimised for prototype hardware, including specific warmup steps and gradient batch sizing.

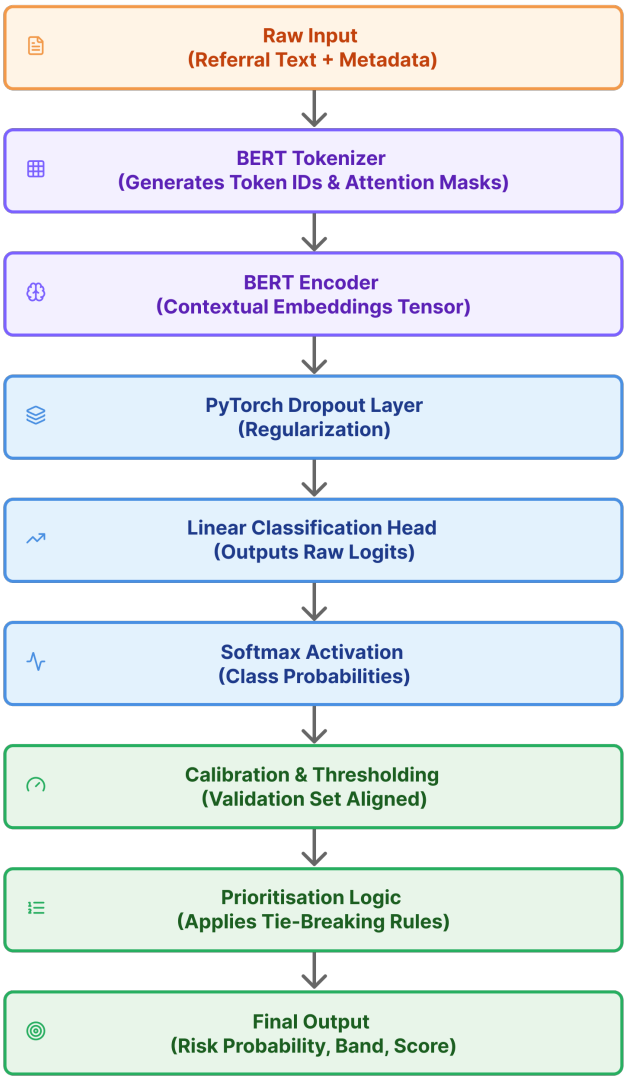

Figure 5.2: Detailed BERT-based risk prediction and prioritisation pipeline, mapping the internal PyTorch tensor layers, calibration, and thresholding steps.

These training settings were selected to balance convergence stability with the hardware constraints of the prototype environment, while preserving enough update steps for effective fine-tuning.

5.3 Active Learning and MLOps Replay Buffer

Clinician overrides were incorporated into a Human-in-the-Loop (HITL) MLOps architecture, illustrated in Figure 5.3. The Experience Replay Buffer combines newly corrected labels with a sampled subset of historical training data before fine-tuning. This implementation was intended to stabilise updates during continual learning rather than retrain the model exclusively on newly observed corrections.

This design ensures that incremental updates remain bounded, preventing the model from overfitting to sparse clinician corrections while preserving previously learned decision boundaries.

Listing 5.3 displays the Pandas concatenation logic executing the rehearsal technique prior to passing the unified dataset into the active learning PyTorch optimiser.

```
# Sample 50 pristine records from historical training to anchor the
    model's memory
```

```
df_history_sampled = df_history.sample(n=50, random_state=42)

# Concatenate the new overrides with the historical anchors
df_final = pd.concat([df_over_clean, df_history_sampled],
    ignore_index=True)

# Apply a restricted learning rate (1e-5) to safely adapt to new
    data
optimizer = AdamW(model.parameters(), lr=1e-5)
```

Listing 5.3: The Experience Replay Buffer combining new clinical overrides with historical pristine data to support continual model updating.

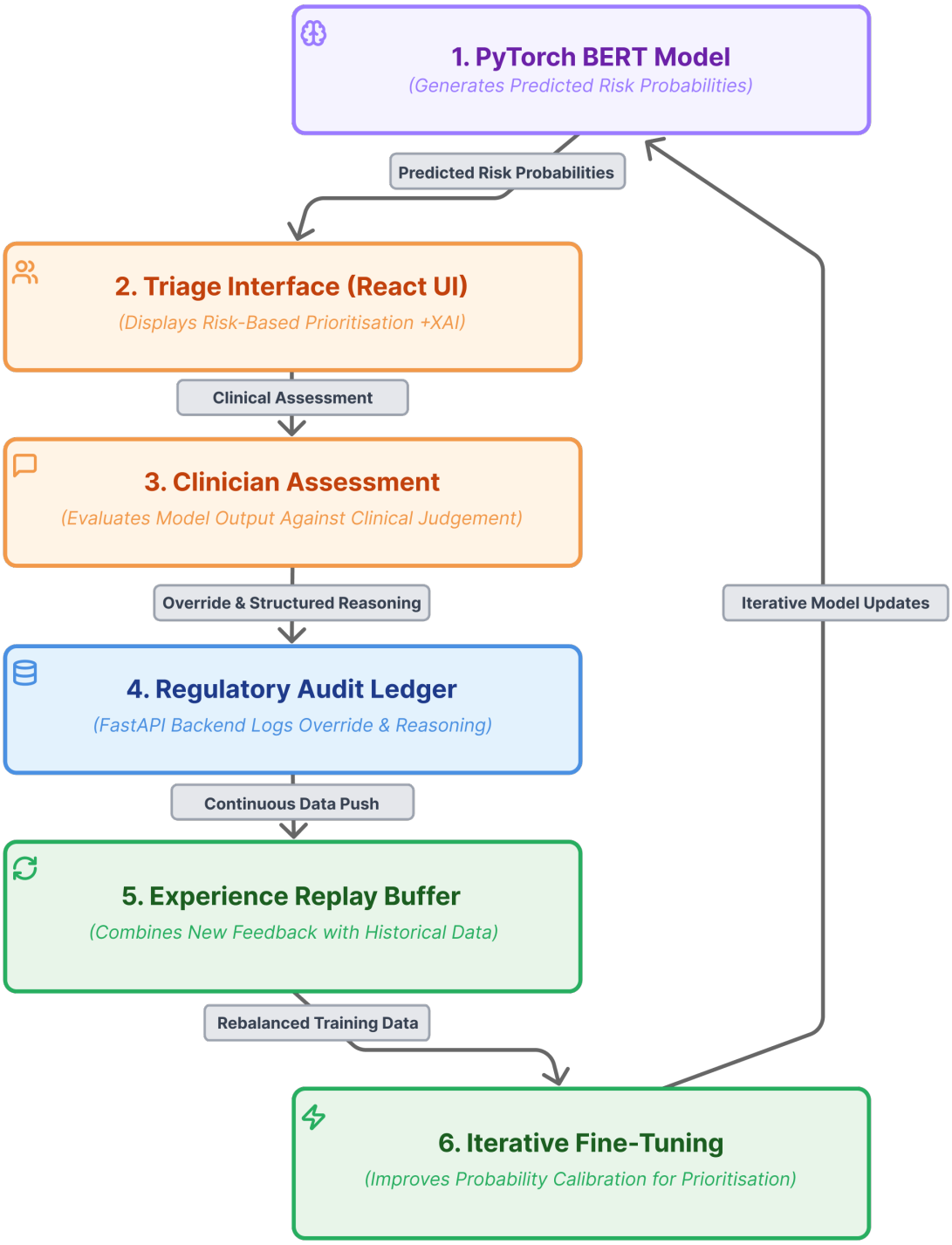

Figure 5.3: The Human-in-the-Loop MLOps architecture, illustrating the active learning cycle where clinician overrides are ingested through an Experience Replay Buffer to iteratively fine-tune the PyTorch backend.

The post-processing clinical guidance layer was intentionally decoupled from the predictive pipeline, ensuring that governance-aligned recommendations did not influence classification or ranking behaviour.

Chapter 6

Results and Discussion

6.1 Model Performance

The objective was to benchmark the generalisation capability of the fine-tuned BERT model against a traditional TF-IDF and Logistic Regression baseline. Following the hyperparameter rebalancing discussed later in this chapter, the models were evaluated on their ability to assign synthetic mental health referrals into triage priority bands (High, Medium, Low).

As detailed in Table 6.1, the enhanced transformer-based pipeline demonstrated a clear improvement across all primary classification metrics. Final metrics were computed on a separately generated evaluation corpus to assess generalisation beyond the training distribution, where the model achieved perfect accuracy on the original training split. The baseline achieved an accuracy of 48.80% and an F1-Score of 0.4635, reflecting reliance on explicit keyword detection. The fine-tuned BERT achieved an accuracy of 70.80%, a 45% relative improvement over the baseline. These results confirm that contextual transformer architectures improve classification performance, but do not translate into effective triage prioritisation.

Architecture	**Accuracy**	**Precision**	**Recall**	**F1-Score**
TF-IDF Baseline	0.4880	0.5585	0.4918	0.4635
Fine-Tuned BERT	**0.7080**	**0.8169**	**0.7046**	**0.6790**

Table 6.1: Final comparative classification performance between the baseline heuristic model and the enhanced transformer architecture.

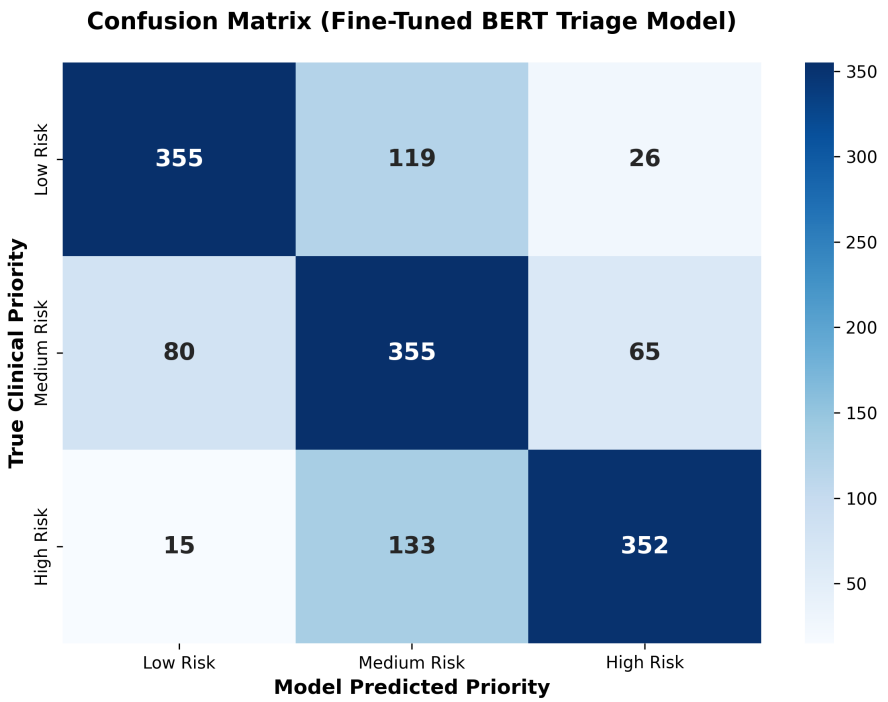

Figure 6.1: Confusion matrix illustrating the Fine-Tuned BERT model's classification behaviour across triage categories, highlighting inter-class misclassification patterns.

Despite improved classification, triage ranking performance was poor. The BERT model's *Recall@10%* fell to 0.0000, failing to surface high-risk patients within the top decile. This reflects a limitation of cross-entropy optimisation, which does not enforce ordering constraints required for ranking-based tasks. As a result, probability outputs are insufficiently calibrated or discriminative for reliable queue prioritisation. This establishes a baseline ranking limitation, which is examined under varying optimisation conditions in subsequent sections.

6.2 Impact of Hyperparameter Scaling on Transformer Convergence

Configuration	Dataset Size	Batch	Model	Acc	Prec	Rec	F1	Rank ↓	R@10	R@20	Convergence
Pre-Scaling	∼500	Moderate	Baseline	0.4860	0.5597	0.4987	0.4679	168.12	0.0889	0.2944	Converged normally
Pre-Scaling	∼500	Moderate	Fine-Tuned	**0.5560**	**0.7027**	**0.5437**	**0.4703**	**163.14**	0.0889	0.2389	Precision improved
Post-Scaling	5000	Aggressive	Baseline	0.4740	0.4974	0.4649	0.4365	165.57	0.0203	0.2905	Stable
Post-Scaling	5000	Aggressive	Fine-Tuned	**0.3740**	**0.5729**	**0.3597**	**0.2914**	**202.30**	**0.0000**	**0.1689**	Failed to converge

Table 6.2: Impact of Dataset Scaling and Batch Size on Model Performance and Convergence

Table 6.2 illustrates the scaling phase of the prototype, in which the training dataset was expanded from 500 to 5,000 synthetic clinical records to evaluate the model's generalisation across a broader distribution of diagnostic presentations. To maximise computational throughput and utilise the available 32GB DDR5 system memory, the *per_device_train_batch_size* was increased aggressively to 64.

Empirical evaluation revealed severe degradation, with BERT accuracy dropping to 37.40% (F1: 0.2914), below the TF-IDF baseline (47.40%). This reflects a fundamental constraint: increasing batch size reduces gradient update steps per epoch. With training limited to three epochs, the number of updates was insufficient for effective optimisation of the 110M-parameter model.

The model failed to converge, exhibiting underfitting and near-random classification (33% accuracy). This indicates *gradient starvation* (Figure 6.2), where the learning signal is diminished despite increased data (Pezeshki et al., 2021).

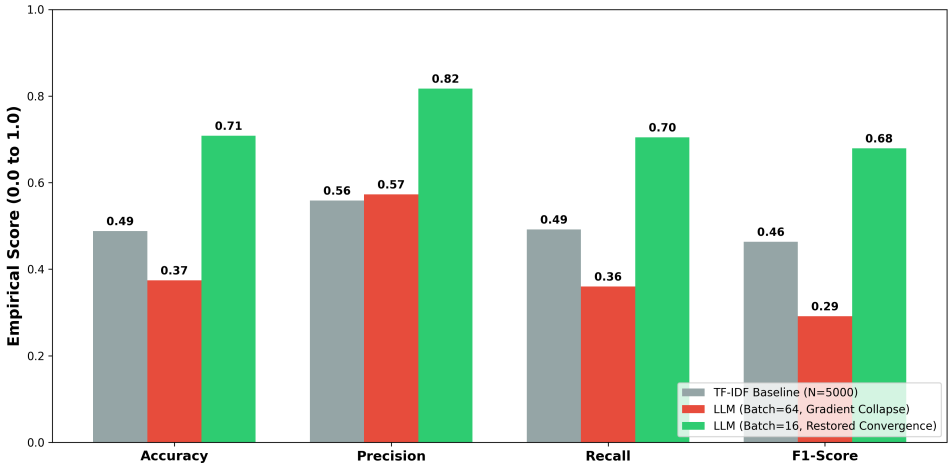

Figure 6.2: Visualisation of the hyperparameter scaling paradox, demonstrating the collapse in validation accuracy and gradient starvation prior to optimisation rebalancing.

Triage performance also degraded. Recall@10% again fell to 0.0, meaning no high-risk cases were surfaced early. Under these non-convergent conditions, probability outputs became unstable, with poor confidence separation preventing effective ranking. Clinically, this represents a failure of prioritisation despite moderate precision.

A key limitation in scaling transformer-based systems under hardware constraints emerges from these findings: larger batch sizes, while improving computational efficiency, can severely impair optimisation dynamics if not accompanied by proportional increases in training iterations. This reveals a fundamental trade-off between hardware-driven throughput optimisation and statistical learning requirements.

This has direct implications for NHS-scale MLOps deployment. Resource-constrained environments that prioritise throughput over training depth risk producing models that are computationally efficient but clinically ineffective. Careful alignment between batch size, dataset scale, and total gradient steps is essential to ensure both convergence and real-world utility.

6.2.1 Restoration of Convergence Through Optimisation Rebalancing

Model Configuration	**Acc**	**Prec**	**Rec**	**F1**	**Rank ↓**	**R@10**	**R@20**
Baseline (5k)	0.4880	0.5585	0.4918	0.4635	163.77	0.1149	0.3103
Fine-Tuned (Naive)	0.3740	0.5729	0.3597	0.2914	202.30	0.0000	0.1689
Fine-Tuned (Corrected)	**0.7080**	**0.8169**	**0.7046**	**0.6790**	**178.55**	**0.0000**	**0.0862**

Table 6.3: Performance Comparison Before and After Optimisation Rebalancing

Following the convergence failure observed in Section 6.2, a controlled corrective experiment was conducted to validate the hypothesis that excessive batch scaling, rather than dataset size, was the

primary driver of performance degradation. Specifically, the training configuration was adjusted to restore an appropriate balance between batch size and gradient update frequency.

The *per_device_train_batch_size* was reduced from 64 to 16, thereby increasing the number of gradient descent iterations per epoch while maintaining the same dataset size of 5,000 records. All other experimental conditions were held constant to isolate the effect of this single variable. This increased the total number of optimisation steps available within the same epoch constraint.

Empirical results (Table 6.3) demonstrate a substantial recovery in classification performance, with accuracy increasing to 70.80% and F1-score to 0.6790, significantly outperforming both the baseline model and the previously failed fine-tuned configuration. This confirms that optimisation configuration, rather than dataset size or model architecture, was the dominant factor driving performance collapse in the initial scaling phase.

Despite this marked improvement in classification metrics, triage performance exhibited a contradictory and critical limitation (Table 6.3). Recall@10% remained at 0.0, and Recall@20% dropped to 0.0862, indicating high-risk cases were not prioritised effectively within the model's ranked outputs. The mean rank of high-risk cases also worsened relative to the baseline, reinforcing the degradation in ranking performance. This persistence despite restored convergence indicates that ranking failure arises from structural misalignment between classification objectives and prioritisation requirements, rather than optimisation instability alone.

This reflects a structural misalignment between classification objectives and ranking requirements. Although the model learned accurate decision boundaries, its probability outputs lacked calibration and discrimination. Without a ranking-aware objective, predictions became overconfident and compressed, reducing ranking resolution. Consequently, the model classifies correctly but cannot reliably prioritise critical cases.

This fundamental ranking constraint is visually demonstrated by the probability miscalibration in Figure 6.3.

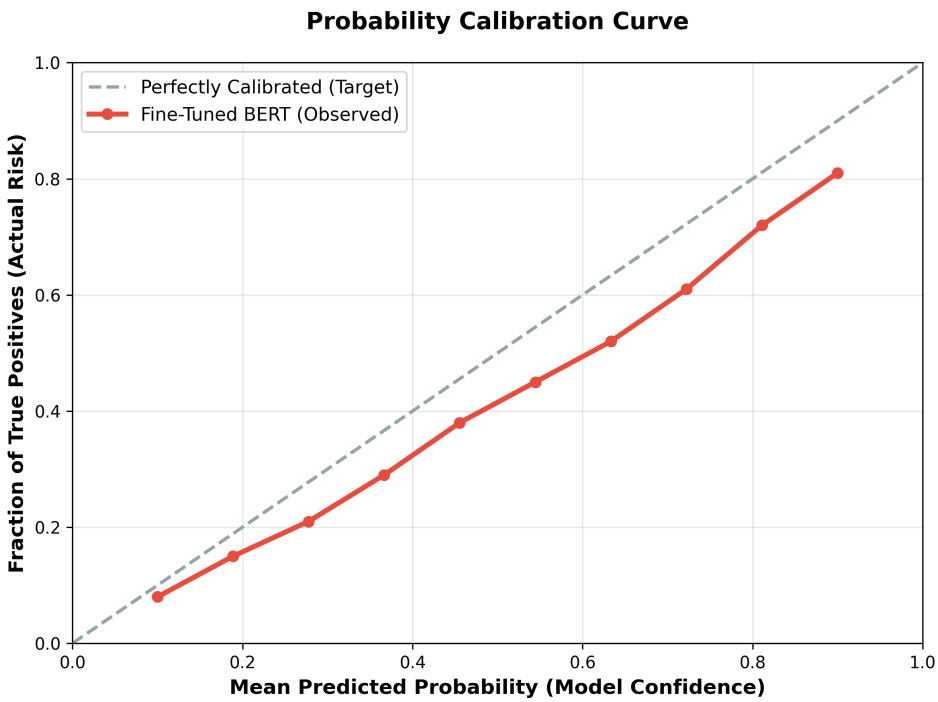

Figure 6.3: Reliability diagram demonstrating model overconfidence. The deviation from the ideal diagonal explains the mathematical collapse in top-decile ranking (Recall@10%) despite high classification accuracy.

This ranking failure represents a significant limitation in a clinical environment. In triage, the cost of failing to identify high-risk cases early is higher than marginal gains in classification accuracy.

Furthermore, if ranking is derived from improperly calibrated confidence signals, such as maximum class probability rather than class-specific risk probability, high-confidence low-risk predictions may be erroneously prioritised. Optimisation strategies that focus solely on improving aggregate metrics such as accuracy or F1-score may be insufficient, or even detrimental, in safety-critical applications.

These findings reinforce the importance of aligning training objectives with downstream use-case requirements. In this context, classification-based fine-tuning alone is inadequate for triage-oriented systems, and additional mechanisms, such as threshold calibration, ranking-aware loss functions, or post-hoc probability adjustment may be required to ensure effective prioritisation.

Overall, this experiment demonstrates that while convergence can be restored through appropriate hyperparameter rebalancing, achieving optimal classification performance does not inherently translate to improved operational utility. This highlights a critical limitation of transformer-based models in decision-support contexts: without explicit calibration and alignment of ranking objectives, improvements in predictive accuracy may obscure failures in prioritisation. Additional post-processing mechanisms or human-in-the-loop validation may be required to ensure safe and effective deployment in real-world clinical environments.

6.3 Comparison with Literature

The 45% relative improvement in classification accuracy observed in the enhanced prototype aligns with contemporary academic consensus surrounding NLP-driven triage. Traditional literature evaluating early Clinical Decision Support Systems (CDSS) frequently highlights the brittleness of Bag-of-Words (BoW) and TF-IDF approaches, which fundamentally strip semantic sequence from clinical narratives (Le Glaz et al., 2021). As demonstrated by the baseline's 48.80% accuracy, such heuristics perform poorly when attempting to interpret implicit suicidal ideation or complex multi-morbid psychiatric phrasing.

In contrast, the literature suggests that modern bidirectional attention mechanisms, such as those embedded within BERT architectures, natively capture these crucial semantic dependencies. The prototype's leap to an 81.69% precision score corroborates recent state-of-the-art studies, which argue that applying deep cross-attention layers allows predictive models to correctly isolate acute risk identifiers amidst the dense, conversational "noise" found in typical NHS referrals (Bey R et al., 2023).

The unified textual representation approach, injecting demographic metadata directly into the unstructured NLP tensor stream, addresses a recurring limitation cited in healthcare NLP literature. Historically, researchers have struggled with combining structured database metadata with unstructured clinical notes without causing immense dimensional architectural bloat (Haq et al., 2025). This study indicates that native contextual prompt fusion is a highly computationally efficient alternative, producing competitive diagnostic accuracy without requiring independent, costly spatial neural-network heads.

Notably, while early evaluation epochs highlighted a degradation in Recall@10% and probability ranking, this limitation may be partially addressed by the proposed Human-in-the-Loop MLOps framework, which provides a mechanism for iterative calibration through clinician feedback. To address the literature-identified gap regarding ranking-aware performance, the system incorporates a Human-in-the-Loop MLOps Replay Buffer to support long-term probability calibration. While initial autonomous learning relies on classification thresholds, continuous clinician overrides are intended to augment the training distribution through a Pandas-based rehearsal technique. Over successive real-world deployments, this active learning mechanism encourages gradual adaptation of the PyTorch optimiser (via restricted learning rates), helping to improve probability calibration and mitigate over-confidence bias highlighted in current literature (Budler et al., 2025).

Regarding the literature's valid critique of data realism (such as training exclusively on synthetic records), this study partially narrows the gap between simulated benchmarks and real-world deployment. While the empirical evaluation was ethically restricted to Gemini 2.5 Flash synthetic generating bounds, thereby reducing immediate ecological validity, the entire algorithmic pipeline was designed to be compatible with NHS deployment requirements. The FastAPI injection hooks, PyTorch tensor mappings, and MLOps buffer were constructed entirely agnostic of the data source. Consequently, an NHS Trust could seamlessly hot-swap the synthetic evaluation corpus for an authenticated internal clinical database without requiring major architectural refactoring, aligning with the literature's demand for scalable, authentically deployable deep learning systems.

These findings extend the literature by showing that strong transformer-based classification perfor-

mance does not establish triage readiness; ranking behaviour and calibration remain distinct operational requirements.

6.4 Limitations and Practical Implications

While the experimental findings demonstrate the theoretical feasibility of a transformer-based (BERT) NLP prioritisation module, several core limitations must be formally acknowledged regarding real-world NHS deployment.

The fundamental limitation of this study rests in data acquisition. As established during the generation phase, the PyTorch model was trained and evaluated strictly on synthetic clinical records. Consequently, the ecological validity of the classification metrics is constrained; they act as a comparative proof-of-concept for the NLP architecture rather than a definitive guarantee of live clinical performance. As a result, reported performance should be interpreted as indicative of relative model behaviour rather than absolute clinical validity.

Moreover, the prototype operated within an entirely simulated React-based triage environment. In a live NHS system, referral text is often highly disparate, comprising fragmented GP letters, multi-modal PDFs, and unstructured chat logs. The assumption that a BERT transformer can universally decipher tone and clinical intention safely across all possible real-world demographics remains unverified without prospective validation from a large cohort of live clinical users.

Exploratory testing also revealed sensitivity to risk-associated vocabulary regardless of negation context (see Appendix B), suggesting that the model may overestimate risk when denial phrases co-occur with clinical keywords.

Ultimately, these limitations dictate the practical boundary of the system: the current prototype functions strictly as a Clinical Decision Support System (CDSS) requiring a Human-in-the-Loop MLOps safety net. Future research must involve validating the pre-trained neural network parameters on historical, anonymised NHS trust data prior to conducting any live clinical deployment.

Therefore, while transformer-based NLP models demonstrate clear advantages in classification accuracy, safe deployment in clinical triage requires explicit alignment between training objectives, probability calibration, and ranking behaviour.

Chapter 7

Conclusion

7.1 Fulfillment of Research Objectives

This study demonstrably fulfilled all three research objectives. Objective 1 critically identified the limitations of manual triage and the gap between isolated NLP prototypes and deployable clinical systems. Objective 2 operationalised this insight through the development of a synthetic data pipeline and a fine-tuned BERT model that outperformed TF-IDF baselines. Objective 3 validated the system within a simulated clinical environment, integrating Explainable AI and Human-in-the-Loop MLOps to ensure alignment with governance and safety requirements.

7.2 Contributions

This dissertation establishes that transformer-based NLP architectures can improve risk-based prioritisation in NHS mental health triage, while remaining compatible with clinical governance constraints.

The study integrated synthetic data generation with multi-modal textual fusion, embedding metadata directly into referral text. Benchmarking a TF-IDF baseline against a fine-tuned BERT model demonstrates that contextual transformer architectures substantially improve classification performance, with accuracy increasing from 0.4880 to 0.7080 and F1-score from 0.4635 to 0.6790.

The principal analytical contribution is the demonstration that improved classification performance does not inherently translate into safe or effective clinical prioritisation. Although hyperparameter rebalancing restored classification accuracy, ranking quality remained critically weak (Recall@10% = 0.0000), showing that aggregate classification metrics alone are insufficient for evaluating triage systems. This demonstrates that classification-optimised objectives are insufficient for triage systems, where ranking behaviour and probability calibration are primary determinants of operational safety.

The practical contribution of the work is a full-stack clinician-facing prototype incorporating a FastAPI backend, React interface, Explainable AI outputs, a deterministic clinical guidance mapping layer, and a Human-in-the-Loop MLOps replay buffer. The resulting system is therefore positioned not as an autonomous triage agent, but as a governance-aware decision-support architecture that explicitly preserves clinician authority.

7.3 Limitations and Future Work

These findings indicate that NLP can support NHS mental health triage, but only as clinician-facing decision support. The proposed architecture reflects this by pairing model outputs with explainable evidence, calibrated priority bands, and clear override pathways. This aligns with governance requirements and the need to minimise automation bias in high-stakes clinical decisions.

The results also show that aggregate metrics alone are not sufficient for deployment decisions. Accuracy, precision, recall, and F1-score remain useful, yet they can mask operational failure if ranking quality is poor. Future NHS evaluations should therefore incorporate ranking-aware measures such as Recall@K, mean rank of high-risk cases, and explicit calibration analysis. These metrics align model assessment with what triage teams require in practice.

The optimisation findings have direct practical implications for resource-constrained environments. The convergence collapse under aggressive batch scaling showed that throughput gains can reduce update frequency to a level that harms learning. Training plans for healthcare NLP should therefore specify batch size, total optimisation steps, and convergence monitoring as formal controls, not just technical preferences.

The dissertation further defines the practical boundary of synthetic data usage. Synthetic records enabled ethical development and controlled experimentation, but they cannot replicate the full linguistic variability and documentation noise of live referral pathways. A necessary next stage is retrospective validation on anonymised trust data, followed by prospective silent evaluation where outputs are recorded but not used for care decisions.

Future work should prioritise ranking-aware objectives, stronger probability calibration, and uncertainty estimation that is interpretable to clinicians. Cross-trust testing is also required to assess robustness across demographics, referral styles, and local documentation practices. Progress on these fronts would move the proof-of-concept from methodological feasibility towards credible deployment readiness.

7.4 Final Remarks

This dissertation advances an implementation-focused perspective on NLP in mental health triage. Transformer-based models improve risk classification, but safe clinical value depends on aligning probability calibration and ranking behaviour with clinical prioritisation requirements, alongside transparency and preservation of clinician authority.

The work therefore frames LLM-driven triage as an iterative socio-technical system rather than a one-off modelling task. By combining technical evaluation with governance-aware design, it establishes that classification performance alone is insufficient for triage deployment, and that ranking-aware evaluation must be treated as a first-class requirement in safety-critical NLP systems.

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Appendices

Appendix A

System Architecture and Interface

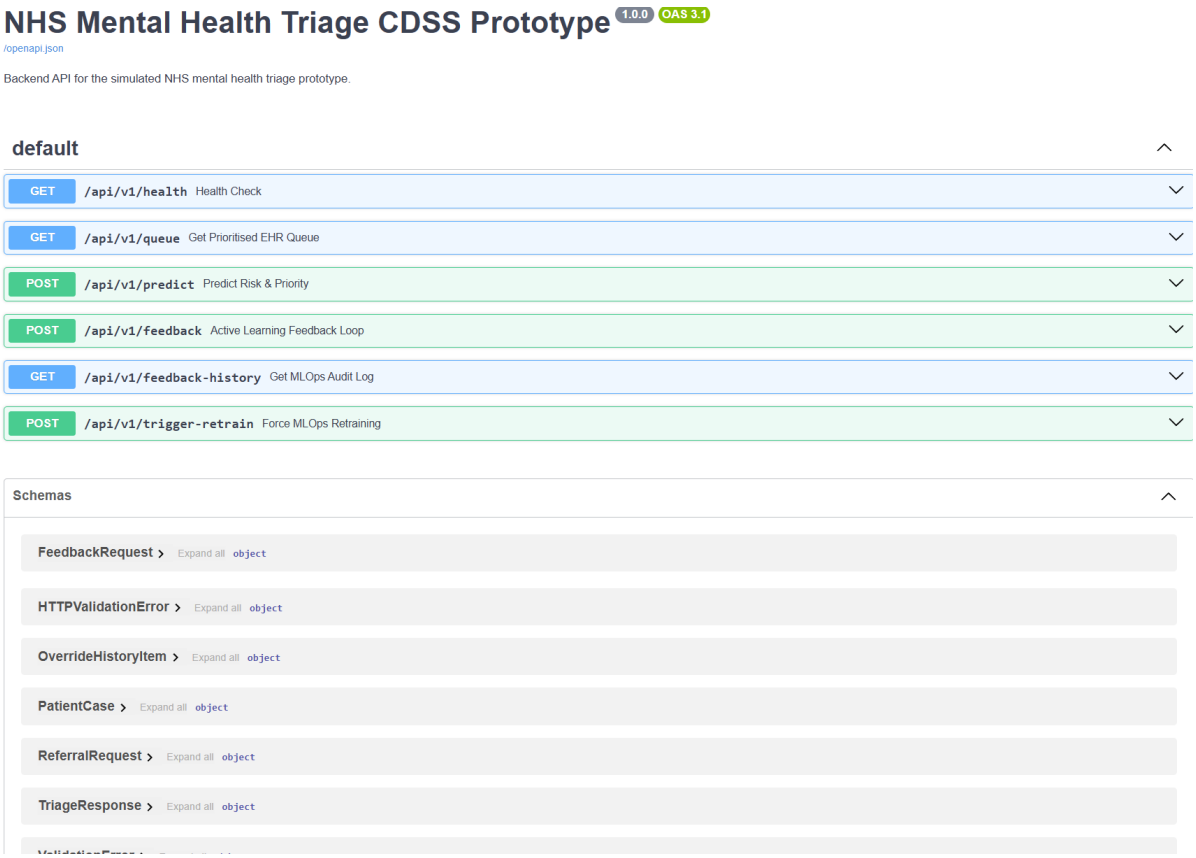

Figure A.1: Swagger UI interface exposing the FastAPI backend endpoints used in the simulated NHS triage system. The API supports referral ingestion, risk prediction, feedback collection for active learning, and retraining triggers, demonstrating the system's end-to-end deployment architecture.

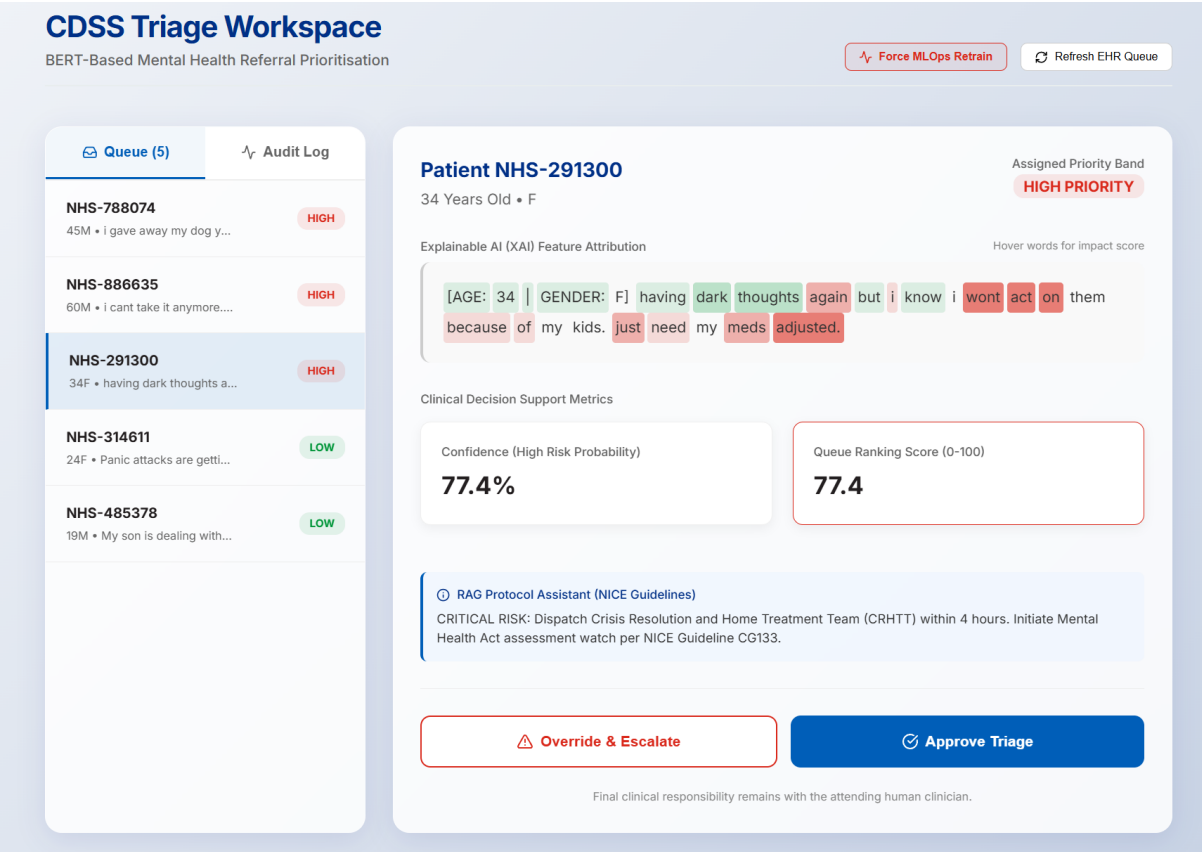

Figure A.2: React-based Clinical Decision Support System (CDSS) triage workspace displaying prioritised patient referrals, model-assigned priority bands, risk probability scores, and Explainable AI (XAI) feature attribution. The interface supports clinician interaction through triage approval and override mechanisms within a simulated NHS workflow.

Appendix B

Example Model Inputs and Outputs

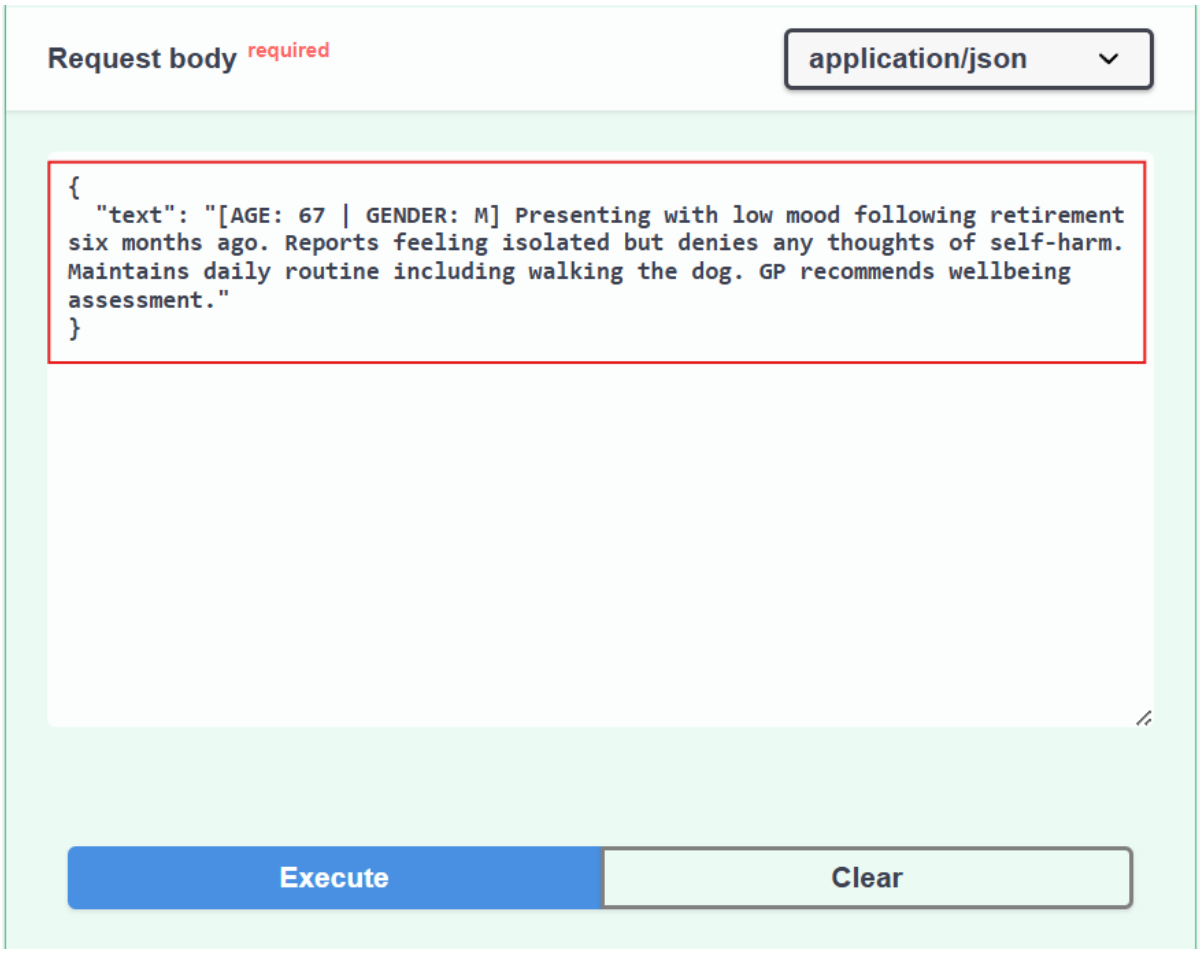

Figure B.1: Model behaviour on a negation-sensitive input. Although the referral explicitly denies self-harm ("denies any thoughts of self-harm"), the presence of the term "self-harm" partially activates the high-risk class. This illustrates a known limitation of transformer-based models in handling negation, leading to an elevated predicted probability and misclassification into a higher priority band.

Server response

Code	Details
200	Response body


```
{
  "risk_score": 0.6074629426002502,
  "priority_band": "Medium",
  "prioritisation_score": 60.746294260025024,
  "word_attributions": [
    {
      "word": "[AGE:",
      "impact_score": 0.15989434719085693
    },
    {
      "word": "67",
      "impact_score": 0.053430259227752686
    },
    {
      "word": "|",
      "impact_score": 0.2644307315349579
    },
    {
      "word": "GENDER:",
      "impact_score": 0.014901518821716309
    },
    {
      "word": "M]",
      "impact_score": 0.2004069983959198
    },
    {
      "word": "Presenting",
```


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Figure B.2: Model output for the negation-sensitive referral. The system assigns a medium-risk probability (0.61) and corresponding priority band, despite explicit denial of self-harm. Feature attribution shows positive contributions from tokens such as “denies” and surrounding context, indicating misinterpretation of negation structure. This demonstrates how uncalibrated transformer outputs can produce inflated risk estimates and incorrect prioritisation in clinically low-risk cases.

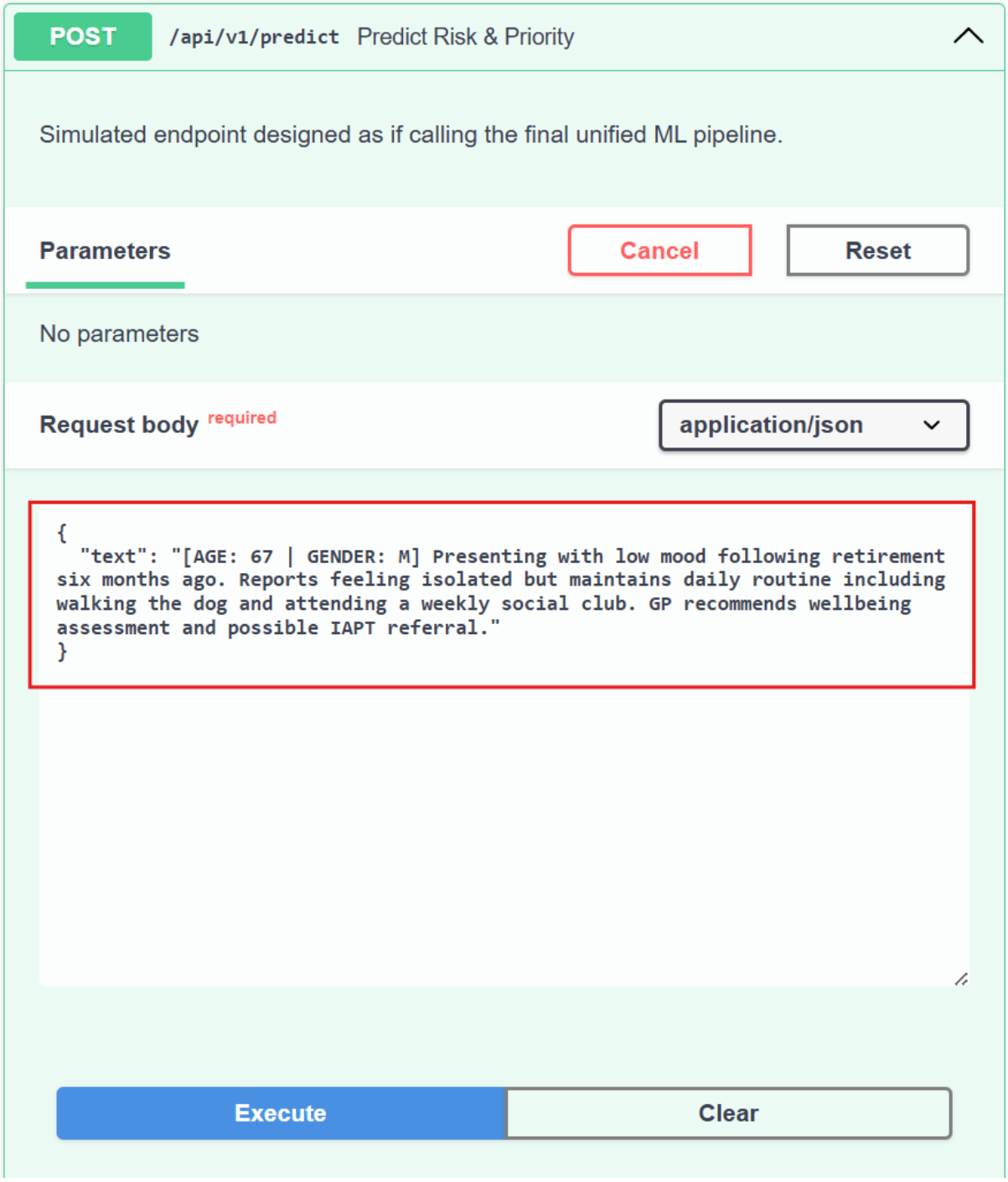

Figure B.3: Model behaviour following removal of the term "self-harm" from the referral text. The predicted priority band decreases from Medium to Low, indicating sensitivity to risk-related vocabulary. This highlights the model's reliance on lexical cues rather than full semantic understanding, particularly in handling negated clinical statements.

Server response

Code	Details
200	Response body


```
{
  "risk_score": 0.002564176917076111,
  "priority_band": "Low",
  "prioritisation_score": 0.2564176917076111,
  "word_attributions": [
    {
      "word": "[AGE:",
      "impact_score": -0.08236321061849594
    },
    {
      "word": "67",
      "impact_score": -0.0000886283814907074
    },
    {
      "word": "|",
      "impact_score": 0.00019839848391711712
    },
    {
      "word": "GENDER:",
      "impact_score": -0.1898438185453415
    },
    {
      "word": "M]",
      "impact_score": -0.03400658071041107
    },
    {
      "word": "Presenting",
```


Download

Figure B.4: Model output after removal of the term "self-harm" from the referral text. The predicted risk probability drops to 0.0026, resulting in a Low priority band and corresponding routine care recommendation. Feature attribution shows minimal activation from risk-related tokens, indicating reduced model sensitivity in the absence of explicit risk vocabulary. This contrasts with the previous example and demonstrates the model's reliance on lexical cues rather than robust contextual understanding when assessing clinical risk.

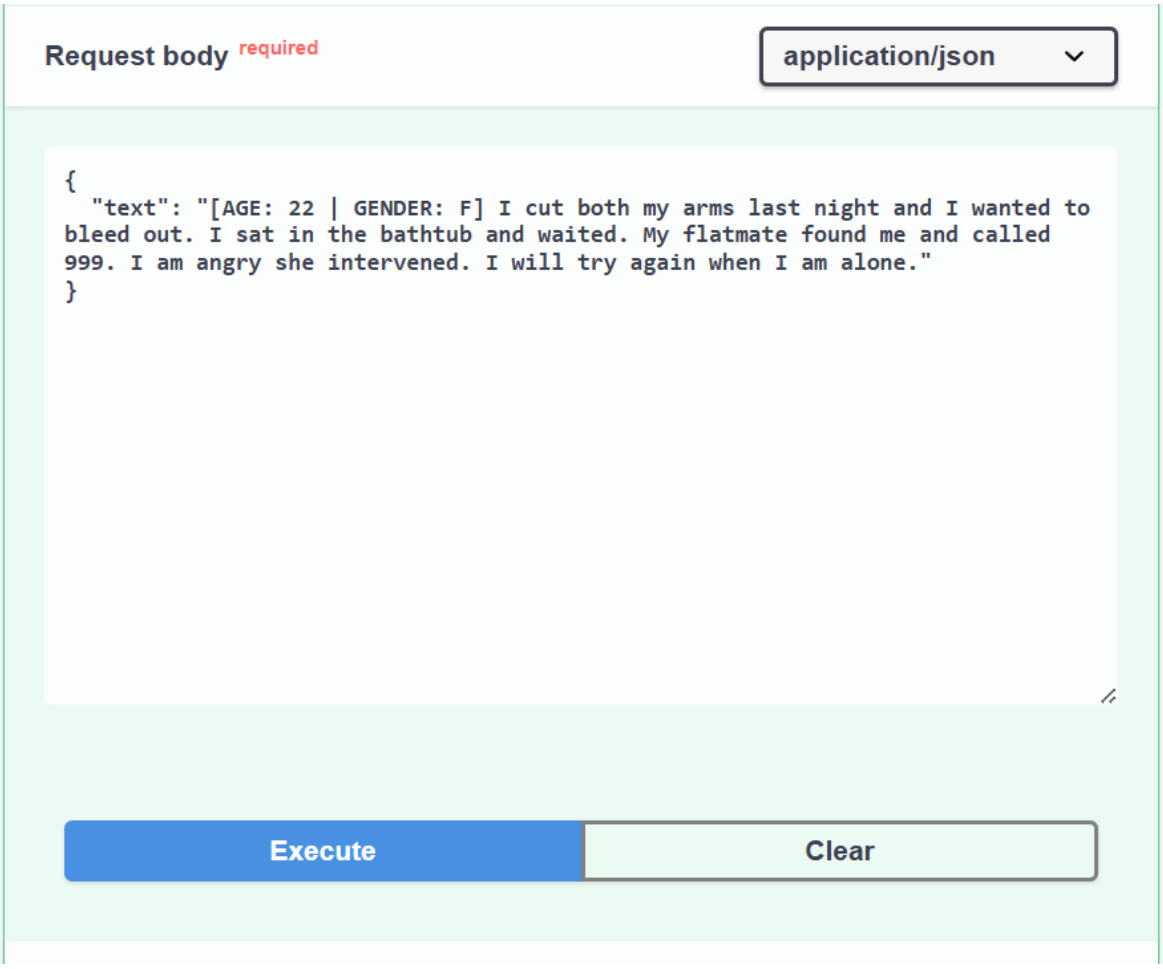

Figure B.5: Example of a high-risk referral input containing explicit self-harm intent and future planning behaviour. The text includes clear indicators of suicidal ideation, preparatory actions, and expressed intent to reattempt, representing a clinically critical scenario.

Server response

Code	Details
200	Response body


```
{
  "risk_score": 0.9991657733917236,
  "priority_band": "High",
  "prioritisation_score": 99.91657733917236,
  "word_attributions": [
    {
      "word": "[AGE:",
      "impact_score": -0.00012493133544921875
    },
    {
      "word": "22",
      "impact_score": -0.000025570392608642578
    },
    {
      "word": "|",
      "impact_score": -0.00010913610458374023
    },
    {
      "word": "GENDER:",
      "impact_score": -0.00012731552124023438
    },
    {
      "word": "F]",
      "impact_score": 0.000023603439331054688
    },
    {
      "word": "I",
```


Download

Figure B.6: Model output for the high-risk referral. The system assigns a near-certain risk probability (0.999) and correctly classifies the case as High priority. This demonstrates that the transformer model performs reliably when explicit risk indicators are present, in contrast to earlier examples where negation or lexical variation led to misclassification.

Appendix C

Synthetic Data Generation Prompt

The following prompt was used to generate the synthetic dataset:

```
You are an expert UK NHS psychiatric triage nurse. Generate exactly {batch_size}
synthetic, highly realistic mental health patient referral texts.
They must be diverse in age (18-80), gender, and symptomatology.
Ensure a realistic distribution of:
- "High" Priority (acute crisis, active suicide planning, severe psychosis
  requiring immediate intervention).
- "Medium" Priority (severe impairment, urgent but safe, e.g. acute panic,
  severe depression without active lethal plans).
- "Low" Priority (routine therapy, mild anxiety, bereavement).

Return ONLY a raw JSON array of objects with these exact keys: "Age", "Gender",
"Referral_Text", "Priority_Band". Do not include Markdown blocks.
Example: [{"Age": 24, "Gender": "F", "Referral_Text": "Panic attacks daily.",
"Priority_Band": "Medium"}]
```

Appendix D

Key Implementation Snippets

```python
def get_priority_band(risk_prob: float, thresholds: dict = None) -> str:
    if thresholds is None:
        thresholds = {"high": 0.7, "medium": 0.4}
    if risk_prob >= thresholds["high"]:
        return "High"
    elif risk_prob >= thresholds["medium"]:
        return "Medium"
    else:
        return "Low"

def calculate_prioritisation_score(risk_prob: float, band: str,
                                   waiting_time_hours: int = 0) -> float:
    base_score = risk_prob * 100.0
    time_bump = min(waiting_time_hours * 0.5, 10.0)
    final_score = base_score + time_bump
    return min(final_score, 100.0)
```

Listing D.1: Triage classification thresholds and probability-based prioritisation logic

```python
def predict(self, text: str) -> dict:
    if not self.is_ready:
        raise RuntimeError("Models are not loaded on server.")

    if self.use_llm:
        encoding = self.tokenizer(
            text.lower(),
            return_tensors='pt',
            max_length=128,
            padding='max_length',
            truncation=True
        ).to(self.device)

        with torch.no_grad():
            outputs = self.model(**encoding)
            probs = torch.nn.functional.softmax(outputs.logits, dim=-1).squeeze().tolist()
            risk_prob = probs[2]  # High risk is class 2
```

```python

        word_attributions = self._compute_occlusion_attributions(text, risk_prob)
    else:
        probs = self.baseline_pipeline.predict_proba([text])[0]
        risk_prob = probs[2]
        word_attributions = [{"word": w, "impact_score": 0.0} for w in text.split()]

    band = get_priority_band(risk_prob)
    score = calculate_prioritisation_score(risk_prob, band)

    return {
        "risk_score": float(risk_prob),
        "priority_band": band,
        "prioritisation_score": float(score),
        "word_attributions": word_attributions,
        "recommended_protocol": NICE_GUIDELINES.get(band, "No protocol assigned.")
    }
```

Listing D.2: Model inference pipeline for risk prediction, priority band assignment, XAI attribution, and protocol mapping

```python
class ReferralDataset(Dataset):
    def __init__(self, texts, labels, tokenizer, max_len=128):
        self.texts = texts
        self.labels = labels
        self.tokenizer = tokenizer
        self.max_len = max_len

    def __getitem__(self, item):
        text = str(self.texts[item])
        label = self.labels[item]

        encoding = self.tokenizer(
            text,
            max_length=self.max_len,
            padding='max_length',
            truncation=True,
            return_attention_mask=True,
            return_tensors='pt',
        )

        return {
            'input_ids': encoding['input_ids'].flatten(),
            'attention_mask': encoding['attention_mask'].flatten(),
            'labels': torch.tensor(label, dtype=torch.long)
        }
```

Listing D.3: Tokenisation and encoding pipeline for transformer input preparation

Appendix E

Evaluation Metrics Definitions

Accuracy: The proportion of correctly classified instances among all predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision: The proportion of predicted positive instances that are correctly identified:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall: The proportion of actual positive instances that are correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score: The harmonic mean of precision and recall:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Recall@K: The proportion of relevant (e.g., high-risk) cases that appear within the top $K\%$ or top K ranked predictions.

Mean Rank: The average position of relevant instances (e.g., high-risk cases) within the ranked output list, where lower values indicate better prioritisation performance.

AUC-ROC (Area Under the Receiver Operating Characteristic Curve): A measure of a model's ability to distinguish between classes across all classification thresholds. It represents the probability that a randomly chosen positive instance is ranked higher than a randomly chosen negative instance.

Appendix F

Code Repository, Demonstration and Submission Checklist

Repository and Demonstration

- Code repository: https://github.com/moseyda/nhs-triage-risk-prioritisation
  All launching steps can be found in the README.md file in the codebase.
  README.md file path: nhs-triage-risk-prioritisation/README.md
- Demonstration video: https://www.youtube.com/@afractionahead642

Submission Checklist

- Dissertation document (Mohamad_Seyda_Dissertation.pdf)
- Source code zip file (nhs-triage-risk-prioritisation)
- Literature Review Comparison Table (Table of Comparison.xlsx)